The FLOPs You Save Are Not the Time You Spend: Score-Then-Select Sparse Attention in Multi-Agent Communication

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Abstract

Sparse attention in multi-agent communication is routinely justified by counting the operations it removes. For rules that select partners by their attention scores, the count is bounded and what it buys in time is smaller still. Because every score is computed before any can be discarded, only the aggregation is removable, so no such rule can remove more than half of attention's arithmetic at any sparsity. Whether even that half becomes time depends on the implementation. Applying the selection as a mask leaves the dense aggregation in place and saves nothing, while a genuinely sparse implementation moves more data than dense attention despite less arithmetic. Fused kernels remove most of that overhead and expose the algorithm: retaining a single peer is faster than the best dense implementation we could obtain, but that case is degenerate, and beyond it the sparse path is one and a half to twenty-eight times slower. Deciding what to attend to before any score exists lifts the bound, and there the selector's own cost decides the outcome. A selector that ranks all pairs in a cheap feature space cannot beat a constant however sparse the attention becomes; a sub-quadratic one is unbounded, and behind a fused kernel its attention stage runs up to sixteen times faster than dense, with the complete path reaching parity at five hundred agents on best-case timings. Count the selector, and measure the kernel that runs.

1 Introduction

Cooperative multi-agent systems increasingly pass learned messages between agents, and attention is the mechanism they use to decide whose message matters (Das et al., 2019; Jiang & Lu, 2018; Jiang et al., 2020). It is a natural choice: the weighting is learned, it adapts to the situation, and it needs no hand-designed communication topology. It is also pairwise, so the channel costs quadratically in the number of agents, and a large part of the literature on learned communication is concerned with restricting who talks to whom.

Restricting communication can mean two different things, and the difference is easy to lose. In a distributed deployment the objective is to reduce the messages physically transmitted, and any restriction achieves that in proportion to the links it removes. In the centralised inference that most of this literature actually runs, the objective is to reduce the computation performed on one device, and there the proportionality does not follow. This paper is about the second. It asks what a particular and natural restriction — keeping, for each agent, the peers with the highest attention scores — saves in local computation, and whether the saving appears as elapsed time.

The efficiency case for that restriction is easy to state and easy to accept. If dense aggregation combines messages from every peer and sparse aggregation combines only a handful, then almost all of the aggregation disappears. What the statement omits is where the handful comes from. Selecting peers by attention score requires the scores, so by the time the selection can be made, every pairwise interaction has been evaluated. Attention has two matrix products of equal size, one producing the scores and one aggregating the values; sparsity of this kind reaches only the second. The arithmetic that can be removed is therefore at most half,

whatever the sparsity, and the familiar fraction that describes the aggregation term alone overstates the achievable saving by a factor approaching two.

An operation count is a model of cost, and it can fail in more than one way. It can describe a computation the implementation does not perform: applying the selection as a mask on the dense attention matrix materialises the discarded entries as zeros and multiplies them anyway, so the executed aggregation is unchanged while a count based on the retained fraction reports a large saving. And the quantity it describes may not govern the outcome: on hardware where time tracks bytes moved rather than arithmetic performed, removing arithmetic from a kernel that is already waiting on memory changes nothing. Both failures occur here, and the second is the one that survives every correction we could make to the first.

The question this paper answers empirically is therefore whether genuinely executing the reduced aggregation makes score-then-select communication faster in the agent-count regime multi-agent communication occupies. We answer it by measuring six implementations of one function, verified equivalent at every configuration whose latency we report, against dense attention through both a library kernel and one we wrote ourselves.

The answer is conditional, and the condition is the contribution. Assembled from general-purpose tensor operations, the sparse path is slower than dense attention everywhere we measure, by up to a factor of thirty-seven, and it moves several times more memory traffic while performing less arithmetic. A fused kernel that keeps the score row in registers and writes no intermediate removes most of that penalty, and where a single peer is retained it is faster than any dense implementation we could obtain, including a search over a hundred tilings of a dense kernel we wrote ourselves. Three qualifications bound that result and we state them with it. The advantage exists only at one retained peer, where the softmax is identically one and no weighting is being learned. It exists only at the narrow head widths where nothing we tried reaches half the device's arithmetic rate; dense efficiency and head width move together in our data and no configuration we ran separates them, so we cannot say which of the two the crossover depends on. And it does not survive an increase in the number of retained peers, because ranking candidates costs more than aggregating over survivors saves. We wrote three selection strategies to test whether that is a property of the ranking or of our implementations of it, including one whose cost is independent of the retained count and whose aggregation is still sparse; none of them changes the outcome.

What follows constructively is the mechanism the negative result identifies. Selection that fixes the retained set before any score exists is not subject to the ceiling, and its operation-count saving is unbounded in principle. Whether it converts into time is a separate question, and we answer it for two selectors the count treats similarly: choosing the nearest peers by distance, which is what neighbourhood-based communication already does, and ordering agents along a space-filling curve, which is sub-quadratic. They remove between three and a hundred and eighty times the arithmetic across the grid we measure, and what they deliver depends on two things the count cannot see. The nearest-neighbour selector ranks a pairwise matrix of its own and reproduces the cost it was introduced to avoid, ending thirty times behind dense attention at the largest agent count. The Z-order selector avoids that, and once the attention over its chosen peers is executed by a kernel written for it rather than composed from library operators — worth four to seventeen times — that attention is up to sixteen times faster than dense. The complete path, selector included, reaches dense attention's cost only at the largest agent count we measure and only on best-case timings; under the median of the same samples it does not.

Contributions.

- We show that any rule selecting communication partners by their attention scores can remove at most half of attention's arithmetic, however sparse it is, and that the bound is the special case of a general accounting in which a selection rule is charged for what it costs to run. That accounting places existing multi-agent restriction mechanisms on one axis and separates those that escape the bound from those that inherit it (Section 3).
- We measure what is actually executed, distinguishing masked, gathered and fused implementations of the same function, and find that the composed sparse path is slower than dense attention throughout the regime by a margin its operation count cannot predict, with memory traffic rather than arithmetic as the cause (Section 4).

- We write fused sparse kernels and a dense baseline searched over one parameter space under one admissibility filter, and report what fraction of the machine each reaches beside the number of configurations each search admitted, so that the comparison identifies the algorithm rather than the authors. Sparsity does become faster than dense attention, at one retained peer, at narrow head widths, and nowhere else (Section 4.3).

- We instantiate the constructive half of the analysis with two selectors that fix the retained set before scoring. Both remove the operation-count bound; one is an order of magnitude slower than dense attention and the other overtakes it. What separates them is whether the selector avoids a quadratic intermediate, and whether the sparse attention left behind is fused. An operation count expresses neither (Section 5).

A secondary pre-registered training study establishes that the architecture being sparsified improves return, so that the sparsification comparisons are not vacuous, and reports six under-powered null contrasts as what they are (Section 6). Every computational claim concerns centralised inference on a single device, not bandwidth-constrained distributed deployment; Section 8 states the full envelope.

2 Related Work

2.1 Communication sparsity in multi-agent reinforcement learning

Learned inter-agent communication has converged on attention as the mechanism for deciding whose message matters. CommNet (Sukhbaatar et al., 2016) broadcasts mean-pooled hidden states; TarMAC (Das et al., 2019) introduces signature-based targeting so that receivers weight senders; MAT (Wen et al., 2022) casts the joint policy as a sequence model. Applied systems inherit the pattern, pairing heterogeneous learners with attention-based message aggregators (Hamdan et al., 2026). Since attention over N agents is pairwise, a large part of this literature also restricts who talks to whom, and the restriction is usually motivated by cost. DGN observes that receiving a large amount of information "requires high bandwidth and incurs high computational complexity" (Jiang et al., 2020), a claim about transmitted bytes and about local compute, of which only the second is what we measure.

Proposition 2 organises these restriction mechanisms by what their selector costs, and the resulting three groups behave differently. The first fixes the retained set from structure that is available before any score exists: DGN restricts attention to each agent's nearest neighbours in the environment (Jiang et al., 2020), ATOC's initiator forms a group from its nearest collaborators within a communication field (Jiang & Lu, 2018), and mean-field methods replace the peer set with a single averaged interaction (Yang et al., 2018). What such a selector costs depends on something the operation count of the attention itself does not record. If the neighbour list arrives with the environment, or is maintained incrementally, or is a fixed topology, the selector is free and the mechanism lands in Corollary 1(d). If instead it is recomputed each step from all pairwise positions — which is how a neighbourhood rule is ordinarily implemented when the agents move — then it is an all-pairs selector in a low-dimensional feature space, case (b), and its saving is capped by a constant however sparse the attention becomes. Section 5 measures the second version and finds the cap binding. The second group scores something cheaper than the full pairwise interaction: IC3Net gates whether an agent speaks at all (Singh et al., 2019), and SchedNet ranks agents by a scalar importance weight and schedules the k largest (Kim et al., 2019). These are top-k rules, but over an $O(N)$ vector rather than an $N \times N$ matrix, so Proposition 1 does not bind them. The third group runs a pairwise selector before running pairwise attention: G2ANet decides which of all N^2 edges survive with a hard-attention network and then applies soft attention to the survivors (Liu et al., 2020b), and MAGIC builds a hard or soft adjacency from an all-pairs interaction before its graph attention (Niu et al., 2021). Their arithmetic saving is real in the aggregation and paid for in the selector, at the same order as the score matrix they avoid.

The mechanism this paper analyses — selecting peers by the attention score itself, after computing all of them — is the limiting case of that third group, and it is rare in multi-agent communication, not absent. The closest published relative is the adaptive sparse communication graph of Sun et al. (2020), which replaces softmax with a generalisation of sparsemax so that the normalisation itself emits exact zeros. It motivates

the design partly through scalability, stating that the resulting sparsity "can reduce the complexity of both the forward pass and the back-propagation of the policy and value networks", and it explicitly considers and rejects hard top-k thresholding as non-differentiable. Because the sparsity-inducing map is applied to logits already formed for every pair, its selection is post-score and sits inside the class Proposition 1 constrains. That places the method in our accounting and does not contradict its claims: its reported gains are in sample efficiency at up to 30 agents and it reports no measured runtime. Handshake-style mechanisms that match a requester's query against every candidate key and then select (Liu et al., 2020a) are score-then-select in the same sense.

The design we analyse is one we proposed and built, so the ceiling constrains our own design and the structural selectors above are outside it. And the agent counts at which these methods operate are small: between two and roughly thirty agents for the attention-based communication methods above, with DGN reaching the low hundreds in grid worlds. Our sweep runs to N=512 so that the trend is visible, but the regime the argument is about is $N \lesssim 10^2$, and beyond it the field has already abandoned all-pairs interaction in favour of neighbourhoods and mean fields — which is to say, in favour of selectors fixed before any score exists.

2.2 Sparse and efficient attention

The efficient-attention literature is where the operation-count argument for sparsity is most developed, and its mechanisms separate along the same axis. Longformer (Beltagy et al., 2020) and BigBird (Zaheer et al., 2020) fix which pairs interact in advance through sliding windows, global tokens and random edges, so the score computation itself becomes sparse and the saving is asymptotic; their selector is essentially free. Reformer (Kitaev et al., 2020) restricts candidate interactions by a locality-sensitive hash and Routing Transformer (Roy et al., 2021) by online k-means clustering; both are content-dependent, both evaluate scores only for candidate pairs, and together they are why the escape proposition carries a selector term at all. Performer (Choromanski et al., 2021) falls outside the premise entirely: it keeps every interaction and reorders the algebra through a kernel factorisation.

Top-k attention in the sequence setting is the closest external analogue of the mechanism we build, and the comparison turns on which efficiency is being claimed. Gupta et al. (2021) process queries in chunks and, for each query, compute its k largest scores against all keys exactly — a score-then-select rule in our sense — and their stated benefit is memory: chunking reduces peak attention memory from $\Omega(L_Q L_K)$ to $O(C L_K)$ for chunk size C while remaining a drop-in replacement in pretrained models. Memory efficiency, operation-count efficiency and wall-clock efficiency are distinct quantities, and a mechanism can improve one at the expense of another; ours is a question about the second and the third.

A related body of work asks how much of the attention function survives truncation, bounding the approximation error of keeping only the largest entries. That question is orthogonal to ours. We hold the function fixed — our gathered, masked and fused top-k implementations agree to floating-point tolerance — and ask whether the sparsification reduces executed computation and elapsed time. A method could be exact and slow, or approximate and fast, and the two literatures answer different halves of the design question.

Sparsemax (Martins & Astudillo, 2016) and α-entmax (Peters et al., 2019) obtain exact zeros from the normalisation rather than a hard cutoff, and make no claim about attention cost; that their zeros arrive after the scores exist, and are therefore subject to our ceiling, is an inference we draw, not a claim they make. Gumbel-softmax (Jang et al., 2017) supplies the relaxation used by learned discrete gates, and constrained-RL formulations (Achiam et al., 2017; Stooke et al., 2020) the machinery for optimising under a communication budget. We implement entmax and a Lagrangian budget as variants in Section 6.

Score-then-select is used at scale in sparse mixture-of-experts, where a router scores every expert and dispatches each token to the best k (Shazeer et al., 2017; Fedus et al., 2022). The structure of the argument is the one we examine, and so is the exemption: these architectures are designed so that the router is small relative to the expert computation it gates, so the analogue of our $N^2 d$ score term is a minor cost, not an equal one, and the saving is real. Whether score-then-select pays therefore depends on the ratio between the cost of scoring all candidates and the cost of the work being skipped — large in mixture-of-experts, and equal to one in single-head attention, which is the ceiling above.

2.3 Hardware-aware sparse execution

That operation counts, memory traffic, kernel fusion and realised latency are distinct quantities is established in the systems literature. The roofline model (Williams et al., 2009) formalises when arithmetic ceases to govern runtime. FlashAttention (Dao et al., 2022) and memory-efficient attention (Rabe & Staats, 2021) make the point from the other direction: the winning implementations of dense attention reduce memory traffic without changing the operation count at all, which is precisely the axis a FLOPs-based argument cannot see. Longformer anticipates the distinction this paper turns on, reporting runtime and memory for three implementations of one attention pattern — a loop version, a vectorised chunked version and a custom CUDA kernel — and noting that the chunked variant consumes roughly twice the optimal memory because it computes some zero values.

Recent systems work establishes where sparse attention does pay, and it bounds our conclusion from the other side. Native Sparse Attention (Yuan et al., 2025) ranks compressed key–value blocks and attends within only the selected blocks, with kernels written for the resulting access pattern, reporting large forward, backward and decoding speedups at 64k context; MiniMax Sparse Attention (Lai et al., 2026) co-designs blockwise selection with its GPU execution path and reports comparable prefill and decoding gains at 1M context. Three properties separate that regime from ours, and all three are load-bearing. Selection is amortised over blocks rather than performed per pair, so the selector cost falls by the block size. Block structure preserves the dense matrix-multiply shape inside each selected tile, so the arithmetic that remains still runs on the hardware path built for it — which per-pair gather, whose aggregation degenerates into a matrix–vector product, does not. And at 10^5–10^6 tokens the skipped computation is enormous, where at $N \lesssim 10^2$ it is bounded by Proposition 1 at a factor of two. We take these results at face value and make no claim of comparability across four orders of magnitude in sequence length; what they identify is what a successor design in our regime would have to change.

Measurement practice. Our training protocol follows the recommendations of Papoudakis et al. (2021) for systematic MARL benchmarking, and we report effect sizes and intervals in place of bare means (Agarwal et al., 2021). Environments come from the PettingZoo/MPE family (Terry et al., 2021) and the learner is MAPPO (Yu et al., 2022). Motivated by Longformer's precedent, we also checked whether the efficiency claims in the papers we drew on are accompanied by a measurement: among the multi-agent papers making such a claim, one of five reports a measured temporal quantity attributable to its mechanism, against five of six among the efficient-attention and mixture-of-experts papers. Twelve papers support no conclusion about either field, and we report the asymmetry only because it characterises the evidence base from which we ourselves adopted the operation-count argument. Appendix D gives the protocol, the completed sheet and its scope caveats.

3 Computational Limits of Score-Then-Select Attention

3.1 Problem setting

Consider one round of scaled dot-product attention over N agents, per batch element, at head width d. Agent i forms a query q_i; every agent contributes a key and a value; self-attention is masked. Write $s_i = q_i K^\top \in \mathbb{R}^N$ for the score vector and $a_i = \mathrm{softmax}(s_i)$ for the attention weights, so that dense attention computes $m_i = a_i V$. A top-k variant keeps the k largest entries of a_i and renormalises over them.

The mechanism this paper analyses is the one that arises most naturally from that description: form the complete score vector, then keep its k largest entries. We call such a rule *score-then-select*, and the term is used in its exact sense throughout. Every analytic claim below concerns that class, not content-dependent selection in general, and Section 3.2 draws the boundary.

Attention has two matrix products of equal size. One produces the pairwise scores; the other aggregates the values. Sparsifying the retained set shrinks the second, because only k value rows are combined. It cannot shrink the first, because the scores are what the selection reads. That observation is the whole of Section 3.2; the rest of this section makes it precise and marks where it stops applying.



Figure 1: Two implementations of dense attention and three of top-k attention, computing two functions in five ways. All five evaluate every pairwise score, because a score-then-select rule must form the complete score vector before it can rank it. They differ in what reaches memory and in how much arithmetic the aggregation performs, and only the gathered and fused top-k variants execute the saving that motivates sparsification at all. The fused row stands for the kernels that compact the retained set; the threshold kernel of Section 4.3 is fused in the same sense but keeps the dense aggregation, so its cost is that of the masked row.

Top-k attention admits three implementations of the identical function, and the distinction between them is what separates an arithmetic saving from an executed one.

Masked top-k. Compute a_i in full, multiply by a $\{0,1\}$ mask, renormalise, and evaluate $m_i = \tilde{a}_i V$ with $\tilde{a}_i$ still dense in memory. The discarded entries are materialised as zeros and multiplied, so the executed aggregation count is unchanged from dense while selection, masking and renormalisation are added on top.

Gathered top-k. Take the top-k of the raw scores, softmax over those k values, gather the corresponding value rows into a (B, N, k, d) tensor, and contract $(B{\cdot}N, 1, k) \times (B{\cdot}N, k, d)$. No zeros are materialised and the aggregation genuinely shrinks.

Fused top-k. Compute the same function in a single kernel, holding the score row in registers, selecting within them, and accumulating the weighted values without writing a score matrix, an index tensor or a gathered value tensor to memory. Section 4.3 reports one.

Gathered and masked top-k compute the same function (Figure 1). Softmax is monotone within a row, so the top-k weights are the top-k scores, and a softmax over just those k scores equals the dense softmax restricted to them and renormalised; the two agree to $< 10^{-10}$ in `float64` (Appendix B). The operation-count argument is about the gathered variant. Timing the masked variant measures an implementation artifact, and timing either against unfused dense alone ignores what a production kernel costs, so we report all implementations against both dense baselines throughout.

3.2 An operation-count ceiling

Analytic counts are multiply–accumulates. Under the convention that a multiply–add is two floating-point operations every term doubles and every ratio is unchanged, so the ceiling is the same number either way; we write MACs where the count is analytic and FLOPs only where a counter attached to the running program produced the figure. Counting the two matrix products that constitute attention, and excluding the shared input and output projections:

$$C_{\text{scores}} = N^2 d, \quad C_{\text{agg}} = N^2 d, \quad C_{\text{dense}} = 2N^2 d, \qquad C_{\text{top-}k} = \underbrace{N^2 d}_{\text{all scores}} + \underbrace{Nkd}_{\text{aggregate } k}. \tag{1}$$

Proposition 1 (Score-then-select ceiling)**.** *Let the retained set be fixed by a score-then-select rule, one whose kept indices are determined only after the complete score vector of each query has been formed. Then*

$$\frac{C_{dense}}{C_{top\text{-}k}} = \frac{2N^2 d}{N^2 d + Nkd} = \frac{2N}{N+k} < 2 \qquad \textit{for all } k \geq 1, \tag{2}$$

with supremum 2 *approached only as* $k/N \to 0$*. Within this class, no choice of* k *or* N *yields more than a twofold reduction in attention MACs.*

Proof. $C_{\text{top-}k} > N^2 d = \frac{1}{2} C_{\text{dense}}$ for every $k \geq 1$ since $Nkd > 0$, so the ratio is below 2. It is decreasing in k, hence largest at $k = 1$, and approaches 2 from below as $k/N \to 0$. □

The familiar figure $1 - k/(N-1)$ describes the value-aggregation term alone. Read as *the* saving it overstates the achievable reduction by a factor approaching two, before any question of whether the reduction is realised in time.

We use $N^2 d$ rather than $N(N-1)d$ because it is the count the implementations execute: self-masking writes $-\infty$ into the diagonal of a full $N \times N$ score matrix and does not skip work. Counting the diagonal exactly gives $2(N-1)/\big((N-1)+k\big)$, which is strictly smaller for every $1 \leq k \leq N-1$, so the proposition is a valid upper bound under either convention.

Five assumptions support the statement, of which only the second is essential. **(A1) Single round**: with R rounds both terms scale by R and the ratio is unchanged. **(A2) Full-score selection**: the retained set is determined only after the complete candidate score vector has been computed, so all N scores must be formed. **(A3) Per head**: with H heads of width d/H both terms scale identically. **(A4) Projections excluded**: including the shared qkv and output projections moves the ratio strictly closer to 1, so the bound remains valid. **(A5) Selection cost excluded**: counting the top-k operation itself can only decrease the ratio, and Section 4.5 shows the term is large in practice.

A2 concerns *when* the retained set is fixed, not whether the rule depends on the data. Mechanisms that are content-dependent but not score-then-select lie outside the class and are unconstrained by Proposition 1: exact or approximate nearest-neighbour retrieval, clustering and routing, indexing and hashing can all identify candidates without materialising the complete pairwise score matrix. Routing Transformer's online k-means routing (Roy et al., 2021) and Reformer's locality-sensitive hash (Kitaev et al., 2020) are content-dependent in every ordinary sense and still evaluate scores only for candidate pairs; a hierarchical rule that ranks compressed blocks and then scores only the selected ones (Yuan et al., 2025) is likewise outside A2. What such rules pay instead is a selector cost, and the next proposition charges them for it.

3.3 Selection before scoring

Proposition 2 (Escape)**.** *Suppose the kept set is determined* without *forming the complete score vector — structurally, or by a retrieval, routing, clustering, indexing or hashing rule — at total selector cost* $C_{\text{sel}}(N, k, d)$*, and that attention scores are then evaluated only for the* Nk *retained pairs. Then*

$$C_{\text{pre}} = 2Nkd + C_{\text{sel}}, \qquad \frac{C_{dense}}{C_{\text{pre}}} = \frac{2N^2 d}{2Nkd + C_{\text{sel}}}. \tag{3}$$

(i) *If* $C_{\text{sel}} = o(Nkd)$ *— in particular for a fixed or precomputed pattern, where* $C_{\text{sel}} \approx 0$*; note that an* $O(N \log N)$ *selector does* not *qualify, since* $N \log N/(Nkd) = \log N/(kd) \not\to 0$ *— the ratio is asymptotic to* N/k*, that is* $\big(C_{dense}/C_{\text{pre}}\big)/(N/k) \to 1$*. Exact equality with* N/k *holds only when* $C_{\text{sel}} = 0$*; for every* $C_{\text{sel}} > 0$ *the ratio is strictly smaller.* (ii) *If only* $C_{\text{sel}} = o(N^2 d)$*, the ratio is no longer bounded by* 2*: with* $k/N \to 0$ *it diverges, though not in general at the rate* N/k*.*

Proof. Both statements read off the displayed ratio. For (i), $2Nkd + C_{\text{sel}} = 2Nkd\,(1 + o(1))$, so $C_{\text{dense}}/C_{\text{pre}} = (N/k)\,(1 + o(1))^{-1}$, whose quotient with N/k tends to 1; substituting $C_{\text{sel}} = 0$ gives $2N^2 d/2Nkd = N/k$ exactly, and any $C_{\text{sel}} > 0$ strictly enlarges the denominator. For (ii), if $k/N \to 0$ then $2Nkd = o(N^2 d)$, so the whole denominator is $o(N^2 d)$ and the ratio diverges. □

Proposition 1 is the case $C_{\text{sel}} \geq N^2 d$. A score-then-select rule must form the full score matrix before it can rank it, so $N^2 d$ is a lower bound on its selector cost and not an estimate of it: the ranking itself is uncounted, and counting it enlarges C_{sel} and shrinks the ratio further. Substituting the bound recovers $2N^2 d/(N^2 d + Nkd)$, which is Proposition 1; any real implementation sits below it. Reading the two propositions as one statement places any peer-selection mechanism on a single axis, the cost of deciding what to keep, and the position on that axis determines what the mechanism can save.

Corollary 1 (What a selector costs, and what it can therefore buy)**.** *Fix* k *and* d *and let* $N \to \infty$ *in* $S = 2N^2 d/(2Nkd + C_{\text{sel}})$*.* (a) Full-score selection*,* $C_{\text{sel}} \geq N^2 d$*:* $S < 2$*.* (b) All-pairs selection in a cheaper feature dimension $p \ll d$*,* $C_{\text{sel}} = \gamma N^2 p$*:* $S \to 2d/(\gamma p)$*, a constant that does not grow with* N *and does not improve with* k*.* (c) Sub-quadratic selection*,* $C_{\text{sel}} = \Theta(N \log N)$*:* $S = \Theta\big(Nd/\log N\big)$*, which diverges, but is a vanishing fraction of* N/k*.* (d) Fixed or precomputed patterns*,* $C_{\text{sel}} = O(Nkd)$ *and in particular* $C_{\text{sel}} \approx 0$*:* $S \sim N/k$*.*

Proof. Each case substitutes its C_{sel} into the denominator. In (b) the $\gamma N^2 p$ term dominates $2Nkd$, giving $S \to 2N^2 d/(\gamma N^2 p)$. In (c) the $\Theta(N \log N)$ term likewise dominates $2Nkd$, giving $S = \Theta(N^2 d/(N \log N))$; the quotient $S/(N/k) = \Theta(kd/\log N) \to 0$. Case (d) is Proposition 2(i) and case (a) is Proposition 1. □

Only (d) delivers the N/k that operation-count arguments for sparse communication usually invoke, and only (a) is bounded. The two intermediate cases are where the multi-agent methods actually sit, and they behave very differently: a selector that ranks all pairs in a cheap feature space escapes the twofold ceiling and still cannot do better than a constant, while a sub-quadratic selector is unbounded but pays a $\log N$ drag that keeps it below N/k by a factor that widens with N. Section 5 measures one selector of each kind. Their analytic ratios are $Nd/(kd + N)$ and $4Nd/(4kd + 30 + 2\lceil \log_2 N \rceil)$ respectively, so at 512 agents and four retained peers the first has reached 14.2 against its ceiling of $d = 16$, while the second is at 107 — unbounded in principle, and already only 72% of the $N/k = 128$ that the naive count promises.

The selector-cost term is material, not decorative. A fixed topology or a precomputed spatial prior really is free at inference and lands in case (i), which is why Longformer and BigBird obtain asymptotic savings where score-then-select top-k cannot (Beltagy et al., 2020; Zaheer et al., 2020): their patterns are fixed before any score is formed, so nothing is spent deciding. A hash must be evaluated and its buckets assembled (Kitaev et al., 2020), and a routing rule must maintain and query its clusters (Roy et al., 2021), so their C_{sel} has to be counted, not assumed away. Between the two sits a family of multi-agent methods whose selector is itself pairwise — a learned hard-attention or scheduling network evaluated over all N^2 pairs before communication attention runs (Liu et al., 2020b; Niu et al., 2021) — and for these C_{sel} is of the same order as the score matrix they avoid, so they inherit a ceiling of the same kind instead of escaping it.

Both propositions concern operation counts only. Whether a pre-score selector attains a proportional *speedup* is a separate question, and one that the measurements in Section 5 answer differently for two selectors that the count treats almost identically.

3.4 Why the count need not govern the time

Proposition 1 bounds a quantity that, in this regime, does not determine runtime. For memory-bound kernels time tracks bytes moved rather than arithmetic performed (Williams et al., 2009), and the composed

implementations of Section 3.1 differ far more in bytes than in operations. Gathering does not remove memory traffic; it relocates it, replacing the multiplication of $N - k$ zeros with the reading, writing and re-reading of a compacted (B, N, k, d) tensor and of the indices that build it.

We compute arithmetic intensity, floating-point operations per byte of global memory traffic, analytically per stage, counting each tensor read and written once, and compare it against a machine balance measured on the same device: a large matrix multiply gives the achievable arithmetic rate, a large copy gives achievable bandwidth, and their ratio is the ridge point above which a kernel is compute-bound. Counting each tensor once ignores cache residency, so at small N, where the working set fits the GPU's L2, the analytic byte counts overstate traffic; the per-stage accounting of Section 4.5 flags such stages so the reader need not infer them. The ridge point itself depends on whether the probe's buffer fits the last-level cache, so we measure two of them per device. On the GPU the ridge is 23 FLOP/byte cache-resident and 53 from memory; on the host it is 8 and 28. Both paths in this study sit below every one of those figures at every agent count, on both devices, so the memory-bound conclusion does not depend on which ridge is the relevant one. The per-stage table of Section 4.5 quotes a third figure, 39, because its probe runs in the same process as the stage timings it scales and measures the arithmetic rate at 57.8 rather than 76.7 TFLOP/s on the same card; every stage sits below that ridge as well. Appendix B.5 gives the probe, including the buffer-initialisation defect that made an earlier version of it overstate host bandwidth several-fold.

Section 4.5 applies this accounting stage by stage and reports, for each stage, how close it runs to whichever machine limit binds it. That is what distinguishes a stage that is badly implemented from one that is doing the only thing its access pattern permits, and the distinction turns out to matter for which parts of the measured penalty a better kernel can remove.

4 Measured Cost

4.1 Method

We time the communication path with the training loop removed. Timing uses on-device CUDA events with an explicit synchronisation on the pinned device, and any diagnostic that transfers a scalar to the host is disabled, because such a transfer synchronises the device inside the timed region. Implementations are timed interleaved repetition by repetition rather than in sequential blocks, so that interference from other tenants lands on all of them at once and their ratios stay paired; the sequential design produced swings above 100% between identical configurations on this shared host. One interleaved group holds every implementation at one agent count, including every retained-peer count of every sparse implementation and the dense implementations they are divided by, so a ratio never spans two groups. The released artifacts record the dense rows with $k = N - 1$, which is a label and not a separate measurement. The primary statistic is the minimum over samples, on the modelling assumption that contention adds time rather than removing it, and the whole timed block is repeated so that the spread of per-run medians is reported and not hidden (Appendix A.2). A minimum estimates an uncontended envelope rather than a service time, so we checked what the choice buys: recomputing every comparison at 192 agents and above from the paired medians of the same samples changes which implementation is faster at one of 24 configurations, 384 agents with a single retained peer, where the ratio moves from 0.94 to 1.14 and the group's recorded inter-run spread is 64%. Neither of the two sweeps that measure the same configuration reproduces the reversal. The pre-score path of Section 5 is the one place where the estimator does change a conclusion, and we report it there. GPU clocks could not be locked; the account lacks the privilege. Operation counts are analytic, from a model that follows the convention `torch.utils.flop_counter` uses and that we checked against it on the composed implementations; the hand-written kernels are outside what that counter observes. Every count in this paper is therefore a model, and where a count and a measurement disagree the measurement governs — which is, after all, the paper's subject.

Two decisions about what is compared deserve stating, because both changed our conclusions. First, we time the attention core — from queries, keys and values to the message tensor — rather than the enclosing module, since the shared input and output projections are identical across implementations and dilute the quantity in question. Second, the dense baseline is the better of PyTorch's fused scaled-dot-product path

Table 1: Forward latency (ms, minimum over interleaved samples) of one function computed six ways, at a quarter of peers retained. The composed sparse path is slower than every dense implementation at every agent count, and the gap widens with N; the fused kernels change that, and Section 4.3 reports where. The last column below N=192 is shown for completeness and supports no claim: those agent counts are launch-bound and do not reproduce (Appendix A.3).

		dense				**top-k**			
N	k	unfused	fused (library)	ours	ours (tuned)	gathered (composed)	fused gather	fused mask	best sparse ÷ best dense
12	2	0.012	0.013	0.004	0.004	0.022	0.004	0.006	0.99
24	4	0.015	0.011	0.004	0.004	0.028	0.006	0.016	1.37
48	8	0.020	0.015	0.008	0.008	0.048	0.012	0.024	1.54
96	16	0.045	0.033	0.018	0.014	0.120	0.049	0.039	2.80
192	32	0.136	0.072	0.036	0.034	0.424	0.387	2.305	11.29
384	64	1.082	0.220	0.119	0.108	2.405	2.750	2.025	18.70
512	128	1.899	0.349	0.188	0.172	6.435	7.214	2.662	15.44

and a dense attention kernel we wrote ourselves in Triton, with the same tiling policy and in the same session as the sparse kernels. At the head widths multi-agent communication uses, the library path is tuned for a different regime, and comparing a hand-written sparse kernel against it credits sparsity with the difference between two authors (Section 7).

Every implementation is checked against the reference at every configuration whose latency any artifact reports — 158 of them, 136 with $k < N - 1$ so that selection is exercised. This is not a formality: exact top-k is discontinuous in the scores, so two implementations that accumulate the score product in different orders can rank a near-tied pair differently and then aggregate over different peers. All agree to `fp32` rounding except on four query rows in 1.25 million, where two scores tie exactly at the k-th position and the threshold-selecting kernel keeps one extra peer (Appendix B).

Unless stated otherwise: PyTorch 2.11 with CUDA 12.8 and Triton 3.6, `fp32`, head width d=16, batch 256, and one NVIDIA Blackwell GPU shared with other tenants.

4.2 What the composed implementations cost

Gathering executes less arithmetic than dense attention (the counter confirms the reduction Proposition 1 permits) and is nonetheless slower than every dense implementation at every agent count we measure (Table 1). Correcting the implementation from masked to gathered helps at small N and stops helping as N grows, because the gather's traffic grows faster than the masked path's wasted arithmetic.

Memory traffic is the mechanism, not arithmetic. Gathering does not remove the score matrix; it adds a compacted (B, N, k, d) tensor and the indices that build it, and then reads both back. Counting bytes gives a criterion that is easier to check than the intensity argument it replaces: gathering is byte-neutral only when the gathered operand is no larger than the score row it avoids reading, that is when $kd \leq N$. At d=16 and N=512 that bound is $k \leq 32$, and the quarter-of-peers operating point is four times past it. The criterion also predicts the reversal in which gathering loses to masking at intermediate N: the gathered operand passes the score row in size before the wasted multiplications catch up.

4.3 What a purpose-built kernel changes

The composed path spends most of its time moving intermediates that a fused kernel need not create. To bound how much of the measured penalty is that composition rather than the algorithm, we wrote fused kernels that hold the score row in registers and never write a score matrix, an index tensor or a gathered value tensor to memory. Two selection strategies are available to such a kernel and they trade against each other. Selecting by k successive argmax-and-mask rounds keeps the aggregation genuinely sparse and makes selection cost proportional to k. Selecting by an exact bisection on the integer key of the score makes

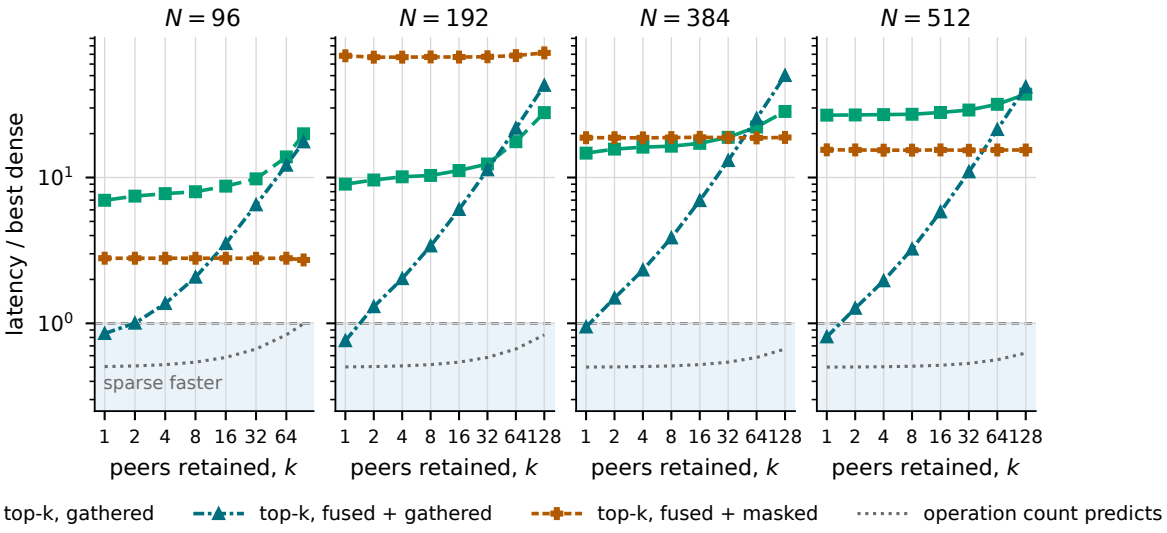


Figure 2: Latency divided by the fastest dense implementation — the vendor library, our own kernel, or our own kernel at the searched tiling — against the number of peers retained. The dotted line is what an operation count predicts for the gathered path, $(N + k)/2N$, which lies below parity everywhere and so predicts a saving at every sparsity. Shading marks the region in which sparsity is actually faster; only the fused gathered kernel enters it, and only at the smallest k. The N=96 panel is shown for continuity and carries no claim, because at that size a fixed per-call cost is an eighth of the measurement and the implementations issue different numbers of calls (Appendix A.3).

Table 2: The sparsity at which the best sparse implementation stops being faster than the best dense one, and how far it is from parity at a quarter of the peers. Both ratio columns are a minimum over the composed path and the two fused kernels that rank in registers, so the widest-k column is sometimes the composed path; the sorted kernel of Table 4 is faster than all three from sixteen retained peers upward and is reported there. Restricted to the agent counts at which the kernels are not launch-bound.

	sparsity that pays		**best ratio**		**at the widest k**	
N	k	of peers	ratio	at k	ratio	at k
192	≤ 1	0.5%	0.76	1	27.9	128
384	≤ 1	0.3%	0.94	1	18.8	128
512	≤ 1	0.2%	0.81	1	15.4	128

selection cost independent of k, but expresses the retained set as a zero weight rather than by compaction, so the aggregation performs dense arithmetic however small k is. Both compute the same function as the composed path.

Fusion removes most of the penalty. At 512 agents retaining a single peer the composed sparse path costs 4.62 ms and the fused one 0.139 ms, a factor of 33 attributable to the implementation and not to the algorithm. At that configuration the fused sparse path is also faster than the best dense implementation available to us.

That statement has to be qualified in turn, and the qualification is the more useful result. Matched authorship is a weaker condition than matched efficiency: two kernels written by the same person in the same session can still differ by more than the margin being measured, and only efficiency against the machine identifies the algorithm. We therefore searched the same parameter space for both kernels under the same admissibility filter, took the better of that search and the vendor library as the dense baseline, and report what fraction of the device's own matrix-multiply rate each kernel reaches beside its latency (Table 3).

Table 3: Both kernels searched over the same parameter space — query tile, key tile or head chunk, warp count and pipeline depth — under the same shared-memory admissibility filter, with the better of the search and the vendor library taken as the dense baseline. The *cfgs* columns are how many configurations each search actually admitted, and they are very unequal: the sparse kernel must hold an N-wide score tile, so above 192 agents only three configurations fit. Efficiency is the executed arithmetic divided by the elapsed time, against the matrix-multiply rate the device attains in the same process.

		best dense				**best sparse**					
d	N	from	cfgs	ms	% peak	k	cfgs	ms	% peak	ratio	k that pays
16	192	triton	106	0.0347	30	1	12	0.0249	21	0.72	≤ 1
16	384	triton	106	0.1082	38	1	3	0.0941	22	0.87	≤ 1
16	512	triton	106	0.1681	44	1	3	0.1293	29	0.77	≤ 1
32	192	triton	94	0.0555	38	1	18	0.0428	24	0.77	≤ 1
32	384	triton	94	0.1923	43	1	3	0.1618	26	0.84	≤ 1
32	512	triton	94	0.3179	47	1	3	0.2231	33	0.70	≤ 2
64	192	sdpa	64	0.0862	48	1	21	0.0962	22	1.12	none
64	384	sdpa	64	0.2720	61	1	3	0.4321	19	1.59	none
64	512	sdpa	64	0.4361	68	1	3	0.5634	26	1.29	none
128	192	sdpa	28	0.1665	50	2	18	0.1724	24	1.04	none
128	384	sdpa	28	0.4688	71	1	3	0.8582	19	1.83	none
128	512	sdpa	28	0.7656	77	1	3	1.1784	25	1.54	none

The search is not equal. All three of our sparse kernels keep the whole score row of a query resident — two rank it in registers and one sorts it in place — so their resident tile grows with the agent count, and above 192 agents the shared-memory filter admits only three configurations against the dense kernel's twenty-eight to a hundred and six. That is a property of the kernels we wrote and not of exact top-k: a selector may compute all N scores without holding them, maintaining a running candidate set as it streams. We did not write one, so the search counts below describe what our implementations admit. We report the counts in the table rather than describing the two searches as equivalent, and the consequence runs in two directions. The sparse kernel's efficiency figure is a lower bound on what a sparse kernel of this shape could reach, so the negative entries at head width 64 and above are weakly supported and we do not build on them. The positive entry is conservative for the same reason: an under-searched sparse kernel that is nonetheless faster is faster.

With that said, two things follow.

The crossover survives tuning, and it is not produced by a sparse kernel that happens to be well written. Across head widths 16 and 32 the sparse kernel is the *less* efficient of the two at every agent count — 21 to 33% of the device's rate against 30 to 47% for dense — and it is faster only because it performs about half the arithmetic.

The crossover is an existence result inside a stated envelope, and the envelope is narrow: one GPU architecture, `fp32`, head widths 16 and 32, agent counts from 192 upward where the sweep reproduces, a single retained peer with one configuration at two, and a dense baseline that reaches under half of the device's arithmetic rate at those head widths. At four retained peers the sparse kernel is already 1.3 to 2.1 times slower, and at a quarter of peers it is more than an order of magnitude slower. The k=1 case is also degenerate as an attention mechanism, because a softmax over one element is identically one and the mechanism reduces to copying the highest-scoring peer's value. What the crossover establishes is that the penalty measured earlier is not irreducible; it is not a recommendation to sparsify.

Why it is confined to narrow heads we cannot say from these measurements. Dense efficiency and head width move together in our data — 30% at d=16 rising to 77% at d=128, as the score product becomes a better-shaped matrix multiply — and no configuration we ran separates them, so "the crossover disappears

Table 4: Three ways of reaching the same retained set, at the two agent counts that carry the crossover. Repeated argmax keeps the aggregation sparse and pays linearly in the number retained; bisecting on the score's bit pattern is flat in it and pays the dense aggregation; sorting score–index pairs once is flat and sparse at the same time, and is not competitive with either at the agent counts where a sort no longer fits alongside a useful block of queries.

		k **argmax rounds**		**bisection**		**one sort**	
N	k	ms	÷dense	ms	÷dense	ms	÷dense
192	1	0.026	0.8	2.392	69.4	0.164	4.7
	2	0.045	1.3	2.435	70.7	0.166	4.8
	4	0.069	2.0	2.435	70.7	0.168	4.9
	8	0.116	3.4	2.435	70.7	0.173	5.0
	16	0.206	6.0	2.441	70.8	0.185	5.4
	32	0.385	11.2	2.440	70.8	0.223	6.5
512	1	0.139	0.8	2.970	17.2	2.433	14.1
	2	0.217	1.3	2.716	15.8	2.450	14.2
	4	0.337	2.0	2.764	16.0	2.493	14.5
	8	0.557	3.2	2.912	16.9	2.490	14.5
	16	1.002	5.8	2.911	16.9	2.316	13.4
	32	1.884	10.9	2.920	16.9	2.651	15.4

once dense attention is near the machine" and "the crossover disappears above d=32" fit our data equally well. Either reading leaves the positive result provisional on a dense kernel we could not write: if one exists at narrow head widths that reaches what the library reaches at wide ones, the crossover goes. Nothing else in the paper depends on it.

Selection is what a successor design has to face, and we tried three ways of doing it. Ranking N candidates to keep k by repeated argmax costs k sequential row-wide reductions, and a reduction of that shape runs at a small fraction of the arithmetic units' throughput because each round depends on the one before it and the hardware has nothing to overlap it with. That is why the argmax kernel's cost grows almost linearly in k while its arithmetic barely does. Bisecting on the score's bit pattern removes the dependence on k, and the only way it can is by expressing the retained set as a zero weight instead of by compaction — so it performs the dense aggregation arithmetic whatever k is.

Whether both can be had at once is the obvious question, and they can (Table 4). Packing each score against its own peer index in one machine word and sorting the row once yields the retained peers already compacted: the aggregation then reads exactly k value rows, while selection costs a single bitonic sort, whose depth grows with the agent count alone and not at all with k. That is visible in what the kernel costs. At 192 agents it runs in 0.164 ms at one retained peer and 0.185 ms at sixteen, a rise of thirteen per cent, against an eightfold rise for the argmax kernel over the same range; what remains of the growth is the aggregation, which is linear in k by construction, and not the selection.

It is also five times the cost of dense attention across that whole range, and it overtakes the argmax kernel only at sixteen retained peers, where both already cost more than five times dense. At 384 and 512 agents it is worse still, thirteen to sixteen times dense at every sparsity below thirty-two retained peers, because a row-wide sort needs the whole score row resident and the block of queries that can share a program shrinks as the row grows. Having both properties does not help. The sort that buys independence from k costs more than the argmax rounds it replaces throughout the range in which sparsity would otherwise pay.

The second obstacle is one dense attention avoids and none of our sparse kernels does. Dense attention streams keys and values in blocks under an online softmax, keeping a small tile resident whatever N is. Each of our sparse kernels instead holds the N-wide score row — in registers to rank it, or in place to sort it — so fewer queries can be in flight per streaming multiprocessor as N grows, and the tiling the search chooses for it is narrower at every size. What is fundamental to exact score-then-select is that all N scores must be *computed* before the retained set is known; that they must all be *resident* is not, and a streaming exact

Table 5: Per-stage accounting for both paths at N=512. The last column is the fraction of whichever machine limit binds the stage, and it separates a stage that is badly implemented from one doing the only thing its access pattern permits. The byte counts are analytic and count each tensor once, so a figure above 100% marks a stage whose working set is partly served from cache. That happens for the softmax stages here and, at smaller agent counts, for the gather; it bounds the model's error at around +50 percentage points, which is larger than several of the gaps in the column and is why we do not read small differences off it. Peak measured on the same device: 57.8 TFLOP/s and 1497 GB/s, a ridge point of 39 FLOP/byte.

stage	GFLOP	GB	FLOP/byte	ms	TFLOP/s	GB/s	% of limit
score product	2.15	0.29	7.53	0.549	3.91	520	35
softmax over N	0.34	0.54	0.62	0.336	1.00	1598	107
dense aggregation	2.15	0.29	7.53	0.235	9.13	1213	81
top-k selection	0.00	0.47	—	3.271	—	143	10
softmax over k	0.08	0.13	0.62	0.059	1.42	2268	151
gather of V	0.00	1.21	—	0.954	—	1265	84
sparse aggregation	0.53	1.14	0.47	0.816	0.65	1398	93

top-k that keeps a running candidate set would relax it. We report the constraint we measured and mark it as ours.

4.4 Sensitivity to head width and batch

Across head widths to 512, the crossover exists only at the two narrowest: the best sparse implementation is faster than the best dense one at head widths 16 and 32 at every agent count, and at no agent count for head width 128 or above, where it is roughly twice as slow (Appendix A). The ratio does not fall monotonically with head width; it reverses.

That is what the byte criterion predicts. Byte-neutrality requires $kd \leq N$, so widening the head at fixed agent count tightens the bound on the retained set proportionally, and at head width 128 the criterion permits only $k \leq 4$ at 512 agents — by which point the selection cost of Section 4.3 has already overtaken the saving. Widening the batch over a thirty-twofold range, by contrast, moves the ratio by less than a fifth and never across parity, which is the expected behaviour of a quantity that is a ratio of per-element costs.

4.5 Where the time goes

Low arithmetic intensity in a batched matrix–vector product is the natural explanation for the sparse contraction's slowness, and it is not the right one. Timed in isolation and charged with its own arithmetic and traffic, both aggregations sit near the bandwidth the device attains on a large copy — 93% for the sparse contraction and 81% for the dense multiply it replaces (Table 5). The gap between them is smaller than the byte model's own error, so the model cannot order the two; what it does establish is that neither sits on a worse hardware path. The sparse one is slower because it moves about four times the bytes, since the compacted operand carries kd elements per query where the score row carries N. The corrected account is simpler, checkable before the code is written, and it is the $kd \leq N$ criterion of Section 4.2.

Selection is a different case, and it is the one stage where the raw decomposition is misleading in the opposite direction. It is the largest stage of the composed sparse path, and it runs at under a tenth of the bandwidth the device attains: it must read the score matrix and write values and indices, and moving those bytes and nothing else would take a small fraction of the time it does. That headroom is real, and the fused kernels realise part of it. What no kernel removes is that the score matrix must be formed and ranked at all, and the fused measurements of Section 4.3 show what remains once the headroom is taken.

4.6 Backward pass and training throughput

Every latency reported so far is a forward pass, which for a training claim is half the question. Timing forward and backward under the same protocol leaves the ordering intact and roughly a third of the margin

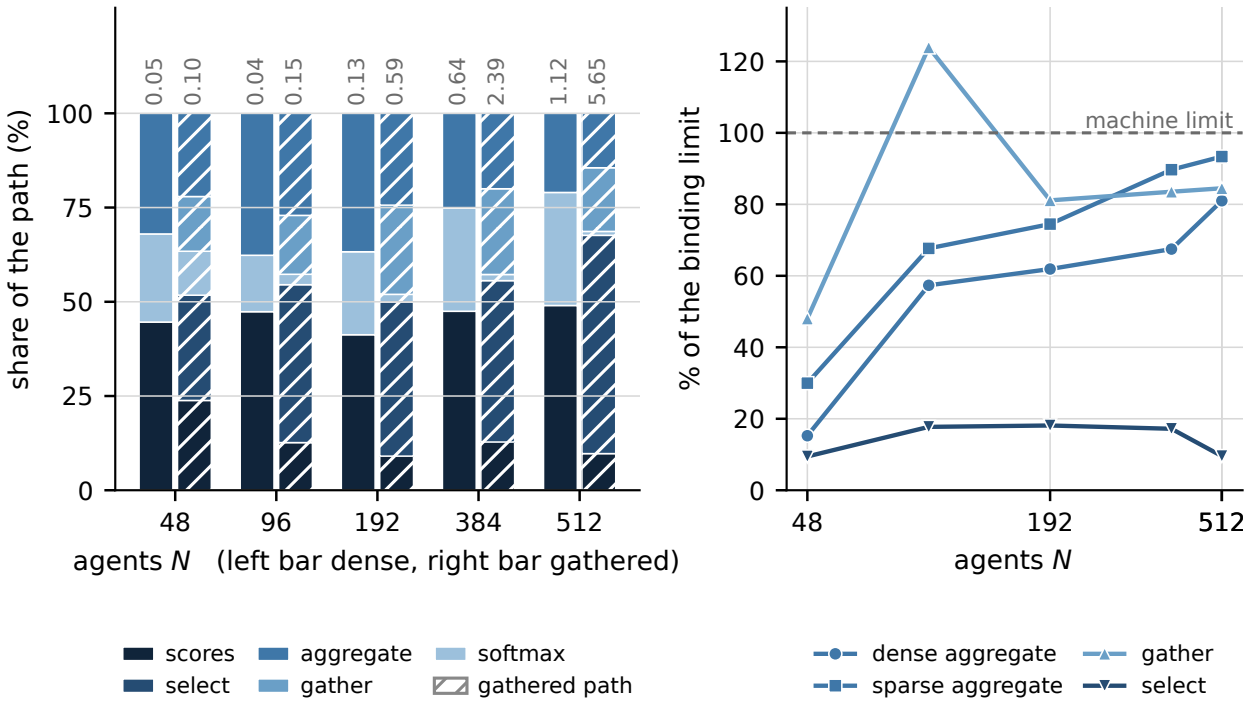


Figure 3: **Left:** how each path divides its time. Selection grows to dominate the gathered path as the agent count rises. **Right:** how close each stage runs to the machine limit that binds it.

Table 6: Forward, backward and their sum, for the differentiable implementations. The sparse path's backward is cheap relative to its forward, because selection has almost no adjoint, so the gap against fused dense narrows without closing. Restricted to the agent counts that reproduce: below 192 the forward and the forward-plus-backward passes are timed in separate interleaved blocks, and the contention between blocks exceeds the backward's own cost, which at two configurations produced a difference a backward pass cannot have.

		dense, unfused			dense, fused			top-k masked			top-k gathered			
N	k	f	b	f+b	f	b	f+b	f	b	f+b	f	b	f+b	gathered f+b ÷ fused f+b
192	32	0.136	0.166	0.302	0.072	0.219	0.291	0.454	0.460	0.914	0.424	0.201	0.625	2.1
384	64	1.082	1.604	2.686	0.220	0.728	0.948	3.065	3.365	6.430	2.405	1.593	3.998	4.2
512	128	1.899	2.648	4.547	0.349	1.203	1.552	6.985	5.748	12.734	6.435	3.142	9.577	6.2

(Table 6). The gathered path's backward costs about half its own forward, because the expensive stage — selection — has almost no adjoint: the retained indices are recorded in the forward pass and the backward is a scatter of gradients into them. Dense attention's backward costs several times its forward. At 512 agents the gathered path is 18.4 times the fused dense path on the forward alone and 6.2 times on forward and backward together, and the same narrowing appears at every agent count in the table. The sparse path's disadvantage in training is therefore substantially smaller than the forward-only measurements suggest, and remains a disadvantage. The fused kernels are excluded from this comparison: they are hand-written forward passes, and timing one against an autograd graph would compare two different things.

End to end, none of this is visible. Timing a complete training iteration by phase, with the communication implementation swapped and everything else fixed, puts the environment and per-step action selection at 78% of the iteration at six agents and 96% at twenty-four, and leaves all five implementations within 2% of one another at every agent count, against an iteration-to-iteration spread of 0.8 to 7.3% (Table 7). The ordering among the five is not stable: the communication-free variant, which performs no attention at all, is

Table 7: Where one MAPPO iteration spends its time. Differences between communication implementations are at or below the iteration-to-iteration spread in the rightmost column.

		share of the iteration (%)							
N	communication	environment	acting	update fwd	update bwd	optimiser	s/iter	÷ fused	spread
6	none	49	29	11	6	4	2.11	1.002	4.6
6	dense, unfused	48	30	12	6	4	2.09	0.992	7.3
6	dense, library fused	48	30	12	6	4	2.10	1.000	4.0
6	top-k, masked	48	30	12	6	4	2.09	0.996	5.4
6	top-k, gathered	49	29	12	6	4	2.13	1.011	5.3
12	none	76	14	5	2	2	4.35	0.998	2.3
12	dense, unfused	75	14	6	4	2	4.39	1.007	3.8
12	dense, library fused	75	14	6	3	2	4.36	1.000	3.3
12	top-k, masked	75	14	6	3	2	4.39	1.008	5.6
12	top-k, gathered	75	14	6	4	2	4.41	1.011	2.3
24	none	92	5	2	1	1	12.84	1.002	4.2
24	dense, unfused	91	5	2	1	1	12.98	1.013	3.3
24	dense, library fused	91	5	2	1	1	12.81	1.000	1.6
24	top-k, masked	91	5	2	1	1	12.80	0.999	0.8
24	top-k, gathered	91	5	2	1	1	12.76	0.996	1.7

fourth of five at six agents and at twenty-four, and the gathered sparse implementation is the fastest of the five at twenty-four. This does not resolve the implementations as equal; it fails to resolve them at all, which is the useful description as well as the honest one. It bounds the practical stakes of the whole measurement campaign in the regime the training study occupies, and it is why the computational argument is made on the module in isolation.

4.7 Robustness

The ordering is unchanged under `torch.compile` and CUDA-graph capture, at `bf16`, and on an x86_64 CPU whose machine balance differs from the GPU's by more than an order of magnitude. The CPU comparison required controlling an effect unrelated to attention: with the default allocator the multi-MiB temporaries each implementation frees between calls are returned to the kernel and re-faulted on the next call, which at N=192 adds roughly 37,000 minor page faults per forward and inflates latency by up to 8.8×. The tax is proportional to the bytes of large temporaries each implementation churns, so it does not cancel in a ratio, and in an un-pinned run it inverted the ordering at one configuration. With the heap pinned the inversion disappears. Appendix B.4 gives the procedure and both runs.

Because clocks are unlocked and the host is shared, we re-ran the identical sweep with the device otherwise idle and again under a concurrent load. With the device idle the ratios reproduce to within a few per cent; under load absolute latency moves by more than 100% while the paired ratios move by roughly half that, and every ratio moves against the sparse path. Pairing removes the typical drift rather than the tail, so the quantity that matters is the margin: no condition we measured brings a composed sparse implementation below parity, and the fused crossover reported in Section 4.3 is measured on the quietest device state we could obtain. Appendix A.3 gives the figures.

5 Selecting Before Scoring

Proposition 2 says that a selector fixing the retained set before any score exists is not bound by the twofold ceiling, and that its operation-count saving grows without limit as the retained set shrinks. That is a statement about counts, and this paper is about what counts fail to predict. This section instantiates the escape and measures it.

We use two selectors, both corresponding to mechanisms multi-agent communication already uses. The first retains each agent's k nearest peers by Euclidean distance over two-dimensional positions, which is what

Table 8: Selecting before scoring, at 2 retained peers, every ratio against the better of the vendor library and our own dense kernel at the tiling the search of Section 4.3 found. Composing the attention stage from library operators costs an order of magnitude more than fusing it, and the difference decides the conclusion: composed, the attention is slower than dense attention at every configuration measured; fused, it is up to seventeen times faster.

		ops	**measured (ms)**			**best**	**÷ best dense**		
				attend			attend		total
N	selector	saved	select	composed	fused	dense	composed	fused	fused
48	nearest neighbours	10	0.049	0.030	0.005	0.008	3.96	0.60	6.91
48	Z-order window	18	0.102	0.031	0.005	0.008	3.99	0.60	13.79
96	nearest neighbours	12	0.100	0.045	0.006	0.014	3.22	0.44	7.58
96	Z-order window	36	0.113	0.045	0.006	0.014	3.23	0.44	8.44
192	nearest neighbours	14	0.324	0.075	0.007	0.034	2.20	0.20	9.51
192	Z-order window	71	0.121	0.077	0.007	0.034	2.24	0.19	3.71
384	nearest neighbours	15	1.757	0.137	0.010	0.108	1.27	0.09	16.42
384	Z-order window	140	0.124	0.138	0.009	0.108	1.28	0.08	1.23
512	nearest neighbours	15	5.449	0.177	0.010	0.172	1.03	0.06	32.41
512	Z-order window	184	0.129	0.177	0.010	0.172	1.03	0.06	0.83

neighbourhood-based communication does (Jiang et al., 2020; Jiang & Lu, 2018). Its selector is still all-pairs, but in two dimensions rather than in the head dimension, so it is Corollary 1(b): its count ratio is $Nd/(kd+N)$, which rises with N towards d and no further, whatever k is. The second orders agents along a Morton curve and retains the k agents adjacent in that order. Sorting is $O(N \log N)$, which is sub-quadratic but not $o(Nkd)$ at fixed k and d, so it is case (c) rather than case (d): its ratio grows without bound, at $\Theta(Nd/\log N)$ rather than N/k. Both then evaluate attention scores for the retained pairs only, so the score term falls from N^2d to Nkd and the ceiling does not apply.

The two are therefore separated before any measurement, by an argument that costs nothing to make and that the operation-count arguments in this literature do not make: one of them cannot exceed a constant and the other can. What the measurements add is that neither the constant nor the divergence settles which is faster, because the count is silent about the execution.

The attention over the retained peers is measured twice: once composed from library operators, and once as a fused kernel that reads one key row and one value row per retained peer and accumulates with a running maximum, writing nothing but the result. Section 4.3 establishes that a composed constant is an upper bound and not a property of a design, and it would be incoherent to apply that standard to the mechanism this paper criticises and not to the one it proposes. The two selectors attend to different peers than a top-k rule would, so what follows compares cost and not quality, and every ratio is against the faster of the library dense kernel and our own.

Fused, the escape delivers, and how much depends on how many peers are kept. At two retained peers the attention stage costs 0.60, 0.44, 0.19, 0.08 and 0.06 times dense attention from 48 to 512 agents; at four, 0.79, 0.46, 0.24, 0.12 and 0.09. The best case over the grid is 0.060, sixteen times faster than dense, at 512 agents and two retained peers. The advantage narrows as k grows and is gone at the smallest agent count by sixteen retained peers, where the stage costs 1.06 times dense behind the Z-order selector and 1.39 behind the nearest-neighbour one, whose retained rows are scattered rather than adjacent. That shape is what the accounting predicts once the score term shrinks with the aggregation, and no score-then-select implementation in this paper comes close to it at any sparsity, because none of them can remove the score term.

Composed from library operators, the same stage is slower than dense attention at every one of the forty configurations, by 1.03 to 5.58 times. Fusing it is worth between 4.0 and 17.3 times, and the factor grows

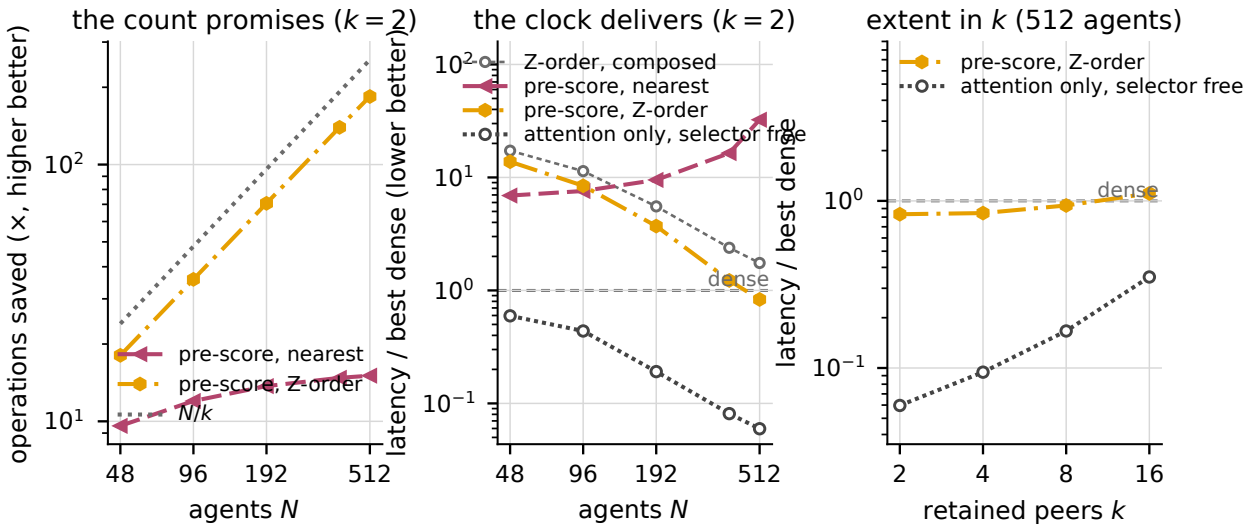


Figure 4: **Left:** the operation-count saving the accounting of Section 3.3 promises. The dotted line is N/k, the figure a free selector would reach; the sub-quadratic selector stays below it by a factor that widens with N, and the all-pairs selector is capped by a constant (Corollary 1). **Middle:** what the same configurations cost. Fused, the attention over pre-chosen peers is more than an order of magnitude faster than dense attention at the largest agent count, and the total is then governed by the selector, which is where the two selectors diverge. **Right:** how far the end-to-end crossing extends. It holds at two and four retained peers, is level with dense at eight and gone by sixteen.

with the agent count because the gathered tensor the composition writes and re-reads grows with it. We report both columns because the composed one is what we measured before writing the kernel, and on its own it supports the opposite conclusion.

End to end the selector decides the outcome, and the two selectors behave oppositely. The Z-order selector's cost is nearly flat in the agent count — 0.102 to 0.142 ms across the range — so its share of the path falls as attention grows more expensive: at four retained peers the whole path costs 13.9, 8.7, 3.8, 1.28 and 0.85 times dense attention from 48 to 512 agents. The nearest-neighbour selector moves the other way. It materialises an $N \times N$ distance matrix and ranks it, which is Corollary 1(b), so beyond 96 agents its cost grows quadratically and its path rises to 32.3 times dense.

The end-to-end crossing at 512 agents needs a qualification the attention stage does not. It is a best-case result and it does not survive a change of estimator. Every latency in this paper is a minimum over interleaved samples, an estimate of what the configuration costs when the shared device is not being disturbed. Under that estimator the Z-order path costs 0.83 times dense at two retained peers and 0.94 at eight, and the two runs of the sweep agree to within 2.3%. Under the paired median of the same samples it costs 2.06 times dense instead. The reason is visible in the dispersion: the pre-score path is several library kernels behind one selector, and on a contended device the tail of a multi-launch path is much heavier than the tail of a single dense kernel, so the median charges it for contention that the minimum removes from both sides. We therefore report the crossing as an uncontended-envelope result and make no claim about end-to-end cost under load. The attention stage carries no such caveat: it is 16.8 times faster than dense on minima and 15.7 on medians.

Three things follow, and together they are the paper's constructive result.

Escaping the operation-count ceiling is necessary and not sufficient. Both selectors escape it, by between 7 and 107 times in operations at four retained peers, and one of them is an order of magnitude slower than dense attention while the other is faster. What separates them is whether the selector reaches its decision through a structure of the size it was introduced to avoid. Corollary 1 says as much before any measurement: the nearest-neighbour selector is capped at a constant by its own all-pairs structure. What the measurement

Table 9: The two positive controls, which were inspected before registration and are excluded from the correction, and the primary pre-registered comparison. $A-B$ is the difference of means and d the standardised effect. Appendix C gives the remaining five comparisons.

	$A-B$	95% interval	d	p
Positive controls, descriptive				
communicating − non-communicating, reference task	+4.26	[+3.15, +5.23]	—	0.0079
communicating − non-communicating, partially observable	+11.59	[+7.13, +16.44]	—	0.0079
Primary pre-registered comparison				
attention-selected peers − uniformly random peers, same budget	+4.03	[−3.35, +11.42]	+0.60	0.357

adds is that the cap binds far below its analytic value, because the selector's cost is traffic rather than arithmetic and the count charges only the arithmetic.

A cheap selector is not sufficient either. With the attention stage composed from library operators the Z-order path is slower than dense at every agent count we measure, and only the fused kernel brings the total to parity at all. Both conditions are needed, and an operation count expresses neither.

And the regime matters, in a way that limits what this result recommends. The crossing appears at 512 agents, the largest count we measure, and we do not extrapolate past it. Most of the multi-agent communication systems we surveyed run with two to thirty agents, and the graph-structured ones reach the low hundreds (Section 2); at those counts our own measurements put the pre-score path behind dense attention, by 13.9 times at 48 agents and 1.28 at 384. So the negative result of Section 4 is the one that bears on current practice, and this section is a demonstration of a mechanism rather than a recommendation to adopt this implementation. What it supplies to a designer is a question with a checkable answer in place of an appeal to an operation count: whether the selector avoids a structure of the size it was introduced to remove, and whether the sparse attention it leaves behind is executed in a kernel written for it.

6 Learning Study

Sparsification would still be worth having if it cost nothing in return. Testing that requires care, because on a task where the channel is not needed "sparsification is free" and "the channel was never used" make identical predictions, so we gate the study on positive controls and register the analysis plan before the outcomes are inspected. The study is secondary, and its role is narrow: to establish that the architecture being sparsified does something, so that the sparsification comparisons are not vacuous, and to report what a five-seed design can and cannot resolve. No claim elsewhere in this paper rests on it.

Twelve variants at up to twelve agents, five seeds each, two million environment steps. Scaling in the number of agents is left to the training-free measurements of Section 4, because a training step in this pipeline is dominated by the environment (Section 4.6) and cannot attribute a latency difference to the communication module. Two positive controls establish that the channel carries something: a reference task in which each agent observes only its partner's goal, fixed at two agents where selection is degenerate, and a partially observable variant of a standard cooperative task, introduced here, in which a peer's relative position is withheld beyond a fixed radius. In the second, the fraction of peers visible falls with the agent count while the absolute number visible rises, because the world does not grow; that confound is why no claim here compares returns across agent counts.

The plan fixes the endpoint, six comparisons and their direction, an exact two-sided permutation test, the effect sizes, a bootstrap interval flagged approximate at five seeds, and a Holm–Bonferroni correction. It also fixes the power expectation in advance: at five seeds a first rejection requires one variant's seeds to lie entirely above the other's, so any null in the family is under-powered and is not evidence of absence, and no equivalence claim is available because no equivalence margin was registered. Appendix C.1 gives the chronology, the arithmetic behind that power statement, and the single amendment.

Both positive controls separate completely, each attaining the floor of the permutation test for this design (Table 9), and measured attention entropy on the partially observable control is 0.35 nats against a ceiling of 1.61, so the learned weighting is peaked rather than uniform. Sparsification is therefore being applied to an architecture whose communication machinery improves return under the evaluated setup. It does not follow that the learned attention channel alone accounts for the difference: the communicating variants carry 7–19% more trainable parameters and the reference task supplies a discrete utterance channel of its own, so the channel-specific causal claim would need a parameter-matched sham variant, which we do not have and do not claim.

The comparison the design exists for holds the communication budget, the executed aggregation cost and every hyperparameter fixed, and varies only whether peers are chosen by attention score or uniformly at random. Its point estimate favours attention by roughly two thirds of a standard deviation and its interval includes zero. The design lacked the power to resolve an effect of that size, which the pre-registered calculation said in advance; nothing follows about whether attention selects informatively. The remaining five comparisons are in Appendix C as point estimates with intervals (Agarwal et al., 2021), and all six are Holm-adjusted nulls.

Two environment limits bound the study further. At twelve agents neither variant learns the task well — the dense variant scores below a uniform-random policy and the sparse variant only just above it — so no comparison at that agent count is interpretable; and the centralised critic sees the same masked observations as the actors, with no privileged global state, which departs from standard practice for this learner and is a plausible contributor.

7 Sources of Measurement Error

Every measurement in this paper was at some point wrong, and in each case the first symptom was a plausible number rather than an obvious error. The causes group into four kinds, and the grouping is useful because the kinds differ in how they are caught. Table 10 lists them with the corrective practice each forced.

Instrumentation. A diagnostic that synchronises the device inside the timed region measures the synchronisation. Our attention module computed an entropy statistic through a call that transfers a scalar to the host, and with it enabled dense latency was nearly flat in N while its work grew by two orders of magnitude — a result we briefly read as evidence that the quadratic cost does not manifest. Synchronising without naming the device is the complementary error: work submitted to a non-default device is never waited on, and every kernel at every size then reports the event-resolution floor of about $0.4\,\mu$s. On the host, the default allocator returns the multi-MiB score matrices each implementation frees between calls, so the next call re-faults them; at N=192 that added roughly 37,000 minor page faults per forward and inflated latency by up to 8.8×. Errors of this kind leave traces — an impossible latency, a curve that is flat where it should be quadratic, a non-monotone cost — and are caught by looking at the shape of the result rather than its value.

Interference from a shared device. Timing implementations in sequential blocks makes each block hostage to whatever else was running during it. On our shared host the recorded inter-run spread reaches 374% at one configuration of the released sweeps, and the sessions that produced our tiling searches differ from each other by up to 81% on identical shapes. Interleaving the implementations repetition by repetition lands any interference burst on all of them at once, so the ratios between them survive even when the absolute numbers drift. The residual sensitivity is to which implementations share an interleaved sweep: at small N, where a fixed per-call cost is a large share of the measurement, the launch count of the whole group enters every measurement in it, and two of our own sweeps that differ only in which implementations they time disagree by 1.4× at twelve agents. Sweeps that will be compared must therefore hold their implementation set fixed and say so.

Measuring a different algorithm from the one claimed. The two most dangerous errors are of this kind, because the resulting numbers are internally consistent, reproducible, and stable under every check the others would fail. They answer a different question from the one being asked. Applying top-k as a

Table 10: Eight ways a benchmark measures something other than the algorithm, each of which produced a spurious result in this study. The first five are caught by the shape of the result; the last three are not, because the numbers they produce are self-consistent.

Kind	Cause	Symptom	Corrective practice
Instrumentation	Host transfer inside the timed region	Dense latency nearly flat in N	Gate diagnostics behind a flag that is off while timing
	Synchronisation without a device argument	Every kernel reports the event-resolution floor	Pin the device and pass it to every synchronisation
	Host allocator returning large temporaries	Up to 8.8× inflation and a non-monotone curve	Pin the heap; count minor faults per forward
Interference	Timing implementations in sequential blocks	Swings above 100% between identical configurations	Interleave repetition by repetition; report paired ratios
	Changing which implementations share a sweep	Two sweeps disagree by 1.4× at twelve agents	Fix the implementation set across comparable sweeps
Wrong algorithm	Benchmarking a masked implementation as sparse	The advertised saving is never executed	Assert executed FLOPs differ and trace the products issued
	Comparing only against unfused dense	A 5.4× more favourable comparison at N=512	Report both baselines; check which backend serves the shape
Wrong baseline	A hand-written sparse kernel against a vendor's dense kernel	An apparent speedup larger than the analytic ceiling permits	Write the dense baseline in the same framework and tiling

mask leaves the dense aggregation matmul in place, so the advertised saving is never executed; a FLOP model based on $1 - k/(N-1)$ nonetheless credits that code with a 75% saving at N=512, k=127, and with more than 90% wherever $k/(N-1) < 0.1$. Comparing against unfused dense alone is the same error in the baseline: at N=512 the unfused and fused dense paths differ by 5.4×, so which one is meant sometimes changes the answer. The defence is to state the operation count each implementation executes alongside its latency, and to check which backend actually serves each shape and dtype, both of which make a mismatch between the claimed and the executed algorithm visible immediately.

Attributing implementation quality to the algorithm. The fourth kind appeared only when we wrote our own kernels. Timing a hand-written sparse kernel against a vendor library's dense kernel measures sparsity and authorship at once. At the head widths used in multi-agent communication, PyTorch's fused attention path runs at roughly half the rate of a straightforward Triton kernel, because it is tuned for the long sequences and wide heads of language models. Against it our fused sparse kernel appears up to 2.8× faster, which is more than sparsity can supply at any k and was how we noticed. Matched authorship is the obvious remedy and it is not sufficient either: two kernels by the same author can differ by more than the margin under test. The condition that identifies the algorithm is matched *efficiency*, so we search one parameter space for both under one admissibility filter — which admits very unequal numbers of configurations, and we report those counts too — and give the fraction of the device's own arithmetic rate each kernel reaches (Section 4.3). That is also what reveals when neither implementation is near the machine, which is a different warning and one worth having: a comparison between two kernels at a third of peak resolves the algorithms only if the margin exceeds the headroom, and ours does not everywhere.

One belief of our own also needed correcting. We assumed that standard counters are blind to fused attention kernels, which would make a FLOPs-only comparison unable to see the baseline that matters. In PyTorch 2.11 that is not so: `torch.utils.flop_counter` reports the fused path correctly, and it was our harness that added an analytic term on top of the counter's output for that implementation, publishing a figure 1.83× too large. The narrower point generalises better than the one we started with: a counter has to be validated

against the kernels it is expected to observe, because only a counter attached to the running program can contradict a FLOP model.

8 Limitations

Every cost result here concerns centralised inference on a single device: the latency and memory of executing the communication module in one process that holds all agents' observations. It is not a claim about bandwidth-constrained distributed deployment, where the objective is to reduce messages physically transmitted and top-k sparsification may well be correct even though it does not reduce local computation. Proposition 1 says nothing about transmitted bytes, and neither do the measurements. The analysis is also single-round and single-head. Assumptions A1 and A3 show that the *operation-count* ratio is invariant to rounds and heads, but this paper's whole argument is that the operation-count ratio does not determine time, so that invariance cannot be used to extend a timing result. It should not be: H heads of width d/H is a change in per-head width, and per-head width is the axis along which our timing result changes sign.

The measured envelope is one GPU architecture (NVIDIA Blackwell, sm_120), one x86_64 CPU, PyTorch 2.11 with CUDA 12.8 and Triton 3.6, input embedding width 64, head width $d \in \{16, \ldots, 512\}$, $N \leq 512$, batch 32–1024, `fp32` and `bf16`, and eager, compiled and CUDA-graph execution. The crossover of Section 4.3 is narrower still: it appears only at $d \in \{16, 32\}$, only at $N \geq 192$, and only in `fp32`, which is the dtype in which dense attention is least advantaged. We could not replicate on a second GPU generation, because the host has two cards of the same model and the hardware that produced this project's earlier runs is gone. Both cards were shared with other tenants throughout. That is why every claim is stated as a ratio between implementations timed in the same interleaved sweep, why the headline sweep was repeated in full, and why no claim is read off an agent count below 192, where fixed per-call cost is an eighth of the measurement.

Our sparse implementations are not optimal and neither is our dense baseline. We searched the same parameter space for both under one admissibility filter, and report what fraction of the device's measured matrix-multiply rate each one reaches, because matched authorship is a weaker condition than matched efficiency and only the second identifies the algorithm. The budgets the filter admits are not equal — three configurations for the sparse kernel above 192 agents against twenty-eight to a hundred and six for the dense one, for the structural reason Section 4.3 gives — so the sparse figure is a lower bound on what a kernel of its shape could reach, and the negative entries at the wider head widths are weakly supported. At the two head widths where the crossover lives no implementation available to us — the vendor library, our own kernel, or any configuration in the search — exceeds 47% of that rate, and the sparse kernel reaches at most 33%. The comparison is therefore between a better dense kernel and a worse sparse one, which bounds the direction of any residual bias but not its size: a dense kernel substantially closer to the machine's limit at this head width would remove the crossover, and we cannot rule one out. Separately, the sparse kernels are forward-only, and they hold the whole score row in registers, which caps them at $N \leq 1024$ and limits how many queries can be in flight at the largest sizes. That cap binds hardest on the sorted kernel, whose sort needs the whole row resident and which is accordingly worst at the largest agent counts. A streaming exact top-k, keeping a running candidate set rather than the row, would not be subject to that cap; we did not write one, so every residency statement in this paper describes our kernels. What no kernel removes is the requirement to *compute* all N scores before the retained set is known, which is what Proposition 1 rests on.

The learning study is secondary and bounded by its power. Five seeds per implementation gives a minimum attainable two-sided permutation p of 0.0079 against a first Holm threshold of 0.0083, so a first rejection requires one implementation's seeds to lie entirely above the other's. Resolving the effect the primary comparison suggests would need roughly 45 seeds per implementation, or 70 at the family-corrected level, from a normal-approximation formula that guides order of magnitude rather than describing the exact test we run. That was a budget allocation and not an unaffordable experiment: the 60-run study consumed 31.9 GPU-hours and a properly powered primary comparison would have cost about 37.9 more. Its positive controls do not isolate the learned channel, for reasons given in Section 6, and one of them sits at an agent count where the selection mechanism is degenerate. Readers should treat the study as evidence that the architecture is not inert, and as no evidence at all about the six comparisons it could not resolve.

Finally, the pre-score selectors of Section 5 demonstrate a cost structure and not a method. They attend to a different peer set than top-k does, so the comparison is of cost and not of quality, and we make no claim that spatial neighbours carry the information attention would have selected. Their selectors are also composed from library operators, so the constant we measure for them is an upper bound in the same way the composed sparse path's was before we fused it; how far a fused selector could fall is the obvious next measurement and one we did not make.

That section's envelope is narrower than the rest of the paper's. It is one head width, one batch and one numerical format — `fp32` at head width 16 and batch 256 — and the same reservation applies as to the crossover of Section 4.3: `fp32` is the format in which dense attention is least advantaged, because our dense kernel forgoes the tensor cores to reproduce the reference's summation order while the pre-score kernel's score is a reduction with no tensor-core path to lose. A format that returns the tensor cores to dense attention would move the crossover, and we did not measure one. The crossover is also at the top of the agent range we measure, so it is stated where it was observed and not extrapolated in either direction.

9 Conclusion

The efficiency case for restricting inter-agent attention to each agent's highest-scoring peers rests on an operation count, and the count is smaller than it looks. Because the selection reads the scores, every pairwise interaction has been evaluated before any is discarded, so only the aggregation is sparsifiable and no choice of sparsity removes more than half of attention's arithmetic. That bound is not a property of top-k but of what the selector costs: written as one accounting in which a rule is charged for deciding, it places every restriction mechanism on a single axis, and the mechanisms that obtain asymptotic savings are exactly those whose selectors are cheap.

Measuring what executes shows the bounded saving to be optimistic twice over. An implementation that applies the selection as a mask performs the dense aggregation anyway, so it executes no saving while a count credits it with a large one. An implementation that genuinely compacts the retained peers executes the saving and is slower regardless, because compaction replaces multiplication by zeros with the writing and re-reading of a gathered tensor, and at these sizes time follows bytes. Gathering is byte-neutral only when the compacted operand is no larger than the score row it avoids re-reading, which is a criterion that can be checked before the code is written.

Whether the remaining penalty belongs to the algorithm or to the tools is a question only kernels can answer, and writing them answered it in both directions. Most of the measured penalty is the composition of general-purpose operators; a fused kernel removes it, and at one retained peer the fused sparse path is faster than any dense implementation we could obtain, including a search over a hundred tilings of a dense kernel of our own. That is the whole of the positive result. It holds at one retained peer, where the softmax is identically one and nothing is being weighted; it holds only at the head widths where no implementation we have reaches half the device's arithmetic rate, and disappears where the vendor library does; and it is gone by four retained peers. At the sparsities we test beyond a single retained peer, the fastest sparse implementation we could write is 1.3 to 28 times slower than dense attention, the penalty growing with the number retained.

The constructive half is where the argument turns, and it turns because the same discipline was applied to it. A selector that fixes the retained set before any score exists is unbound by the ceiling, and the two we measured reduce the operation count by up to two orders of magnitude. Composed from library operators, both are slower than dense attention and we would have reported the escape as a failure. Written as a fused kernel — one key row and one value row read per retained peer, a running maximum, nothing else written — the same attention is up to sixteen times faster than the best dense implementation we have, and improves with the agent count exactly as the accounting predicts.

What then decides the outcome is the selector, and the two we tried diverge. One reaches its decision by ranking a pairwise matrix of its own, reproducing the cost it was introduced to avoid, and its path falls further behind as the agent count grows. The other is sub-quadratic and flat in cost, and at the largest agent count we measure its path reaches parity: retaining two or four peers of five hundred costs about

fifteen per cent less than the best dense attention we can obtain on best-case timings, a margin the repeated sweep reproduces to within a few per cent but which the median of the same samples reverses, because a multi-launch path has the heavier tail on a shared device. We report the attention stage as the robust result and the end-to-end crossing as an uncontended-envelope one. Neither condition is sufficient alone: a cheap selector with a composed attention stage loses, and a fused attention stage behind a quadratic selector loses.

That is the finding we would put in front of a designer. An operation count prices work in operations, and on this hardware, at these sizes, the price is set by the shape the work is executed in. A successor method has to answer for what its selector costs, for whether it reaches its decision without materialising the structure it meant to avoid, and for whether what it leaves behind is executed by a kernel written for it. Only the first is visible in a count, and each of the other two was, in our own measurements, the difference between a design that works and one that does not.

Broader Impact Statement

This work is a measurement and correction study of a computational-efficiency claim. It introduces no new capability and no new data. Its intended effect is to reduce wasted compute in the design of multi-agent communication architectures by discouraging optimisation against an operation count that does not predict runtime.

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A Additional benchmark results

A.1 The full parameter grid

Table 11 gives the best-over-implementations sparse latency divided by the best-over-implementations dense latency, across head width and across batch, at each agent count and at the most favourable sparsity available. Head width is swept to 512 and batch from 32 to 1024. The largest batch we attempted, 4096 at 512 agents, could not be measured: the score matrix alone runs to several gigabytes and the device is shared, so that configuration is absent.

Table 11: Best sparse divided by best dense, over head width (upper block) and batch (lower block). Values below one indicate sparsity winning. The 96-agent column is included for completeness and carries no claim, for the reason given in Appendix A.3. Each column comes from the sweep that varies that axis, so a figure here can differ from the same nominal quantity in Section 4.3 by the few per cent that separates two sessions.

head width d	N=96	N=192	N=512
16	0.85	0.76	0.82
32	0.51	0.79	0.70
64	0.88	1.29	2.23
128	1.41	2.19	1.84
256	2.07	2.04	2.05
512	1.54	1.71	2.31

batch B	N=96	N=192	N=512
32	0.54	0.65	0.54
1024	0.57	0.75	0.72

A.2 Dispersion behind the headline numbers

The primary statistic is the minimum over interleaved samples, on the modelling assumption that contention can only add time. That statistic is trustworthy only if the underlying sample was not wildly dispersed, so the harness records the median, the interquartile range and the spread of per-run medians for every configuration, and the released artifacts carry all of them. Two properties of this host limit what any of it can mean. Both GPUs carried other tenants' allocations and near-continuous utilisation throughout the campaign — tens of gibibytes, varying between sessions, which we observed but did not record per run — and clocks could not be locked. We therefore report the spread of per-run medians beside every configuration rather than a confidence interval: a ratio of minima has no sampling distribution.

A.3 Drift control

The headline sweep was repeated in full on the same device, hours later and under different load from other tenants. Table 12 gives how far each agent count moved between the two runs, in absolute latency and in the ratio against the best dense implementation, which is the quantity every claim is stated in.

Across 44 configurations at each agent count the median movement in latency is 0.1% at 96 agents, 0.2% at 192, 0.6% at 384 and 6.1% at 512, and in the ratio 0.3, 0.2, 0.5 and 7.2%; the worst single configuration moves 12.6%. Which implementation is faster changes between the runs at exactly one configuration, 96 agents with two retained peers, where the ratio is 1.002 in one run and 0.998 in the other — a tie either way.

Two things follow for how the sweeps are read. The measurement reproduces well enough that every ordering in this paper survives repetition, which is the property the claims depend on. And the largest agent count is the least stable one, not the most, so a margin of a few per cent there — which is what the end-to-end crossing of Section 5 is — is at the edge of what this device can resolve, and we say so where we report it. We restrict the crossover of Section 4.3 to 192 agents and above for a different reason: fixed per-call cost. A

Table 12: Movement between two runs of the identical sweep, hours apart under different load. Median movement is at or below one per cent everywhere except 512 agents, where it is six to seven per cent; no single configuration moves by more than 12.6%, and no ordering the paper reports changes between the runs.

		latency $\lvert\Delta\rvert$ (%)		**ratio** $\lvert\Delta\rvert$ (%)	
N	configurations	median	max	median	max
96	44	0.1	4.2	0.3	2.5
192	44	0.2	6.1	0.2	6.0
384	44	0.7	11.3	0.5	11.6
512	44	6.1	12.0	7.2	12.6

one-program Triton kernel that writes a single float, timed the way every measurement here is timed, costs 1.8 µs at its minimum and 2.3 µs at its median (`scripts/launch_overhead.py`). At 96 agents the fastest dense configuration is 13.8 µs and the fastest fused sparse one 11.7 µs, so that fixed cost is an eighth of each, and the composed sparse path issues several kernels rather than one. At 512 agents the same cost is one per cent of the dense measurement. A ratio between implementations that issue different numbers of kernels is therefore partly a ratio of launch counts at the small agent counts, and stops being one by 192.

The pre-score sweep of Section 5 is repeated in the same way, and for a stronger reason: the crossing it reports is a margin of about ten per cent rather than a factor, and a margin of that size has to be shown to survive a second session before it can be reported at all. It does. The median movement in the ratio between the two runs is 0.7% and the worst is 8.5%; at the agent count that carries the crossing the median is 0.8% and the worst 3.6%, and the four cells the crossing itself is read off agree to within 1.5%. Unlike the core sweep, this one reproduces at every agent count it measures, which is why the section quotes it from 48 agents upward.

The repeat is on the same physical card, so it establishes stability across runs and hours, not independence from one device, and we were not able to obtain a second card free of other tenants. The tiling search of Table 3 was run twice, once on each card of the host. The two searches select the same configuration at 63 of the 72 shapes they share, and at head width 16, where most of what this paper reports lives, at 16 of the 18 shapes; the two disagreements are both sparse-kernel tilings. Their absolute times differ by as much as 81% with the contention each session carried. That is why the search contributes a configuration and not a latency: every sweep re-times that configuration in its own interleaved group, and the only place a latency from the search's own session appears is Table 3, whose columns are the search's own record and are labelled as such. All measurements reported here were taken on one of the two cards of this host, and no cross-device calibration is claimed.

B Implementation and measurement validation

B.1 Functional equivalence

Every implementation is checked against the reference at every configuration whose latency any released artifact reports: 158 combinations of agent count, head width and retained-peer count, 136 of them with $k < N - 1$ so that selection is actually exercised, over 1.25 million query rows in total. The check matters more than it looks. Exact top-k is discontinuous in the scores, so two implementations that accumulate the score product in different orders can rank a near-tied pair differently and aggregate over different peers, producing a large output difference from a rounding-level score difference. An earlier version of our per-query fused kernel computed the score row as a reduction instead of through the matrix-multiply units and disagreed with the reference at a handful of configurations for exactly that reason; the kernels reported here use the same product as the reference.

With that fixed, every implementation agrees with the reference to `fp32` rounding — maximum deviation 3.2×10^{-6}, and zero for the two implementations that share the reference's operator sequence exactly — with one attributable exception, which is an ambiguity in the function rather than an error in an implementation.

Exact top-k is only well defined when the k-th largest score is unique. The kernel that selects by threshold retains every peer at or above that score, so a tie there gives it $k{+}1$ peers; the kernel that sorts score–index pairs breaks the same tie by peer index, which is not the order a rank-based selection uses. Four query rows in 1.25 million have such a tie, at three of the 158 configurations, and those rows are the only place any implementation differs from the reference by more than rounding. Two further rows appear in the head-width sweep, at 192 agents and head width 128, where the same two kernels differ from the reference by 1.7 and 1.1 on one query row of 6144 at each of two retained-peer counts; the k-th score on those rows is tied to the bit, so the divergence is the ambiguity and not an error. The check is run by `scripts/verify_all_configurations.py` and by the sweep harness itself, both of which count how many query rows diverge as well as by how much, and admit a tie as the explanation only when no more rows diverge than have a tie.

The fused kernel for attention over a pre-chosen peer set is checked the same way against the composed path it replaces, at all 40 configurations of Section 5 and both selectors, over 315,392 query rows, to a maximum deviation of 1.1×10^{-6}. Its timed call also converts the retained indices to the integer width the kernel reads them at, which the composed path does not need, so the figure we report for it includes a cost the comparison does not require of it.

B.2 What each implementation executes

The masked and gathered implementations compute the same function while executing different work, which is the distinction the paper turns on, so regression tests pin it. Maximum absolute deviation between the two paths is below 10^{-10} in `float64`. A FLOP counter attached to the running program reports identical totals for dense and masked — the counter scores matrix products, and masked issues the same two as dense — against a smaller total for gathered, the difference being exactly the aggregation term. Tracing every matrix product confirms that the gathered path issues no dense aggregation product and does issue the compacted one, and that every top-k variant still issues the full score product, which is assumption A2 made mechanical.

The four hand-written kernels differ only in how the retained set is reached, and each one is stated here in the terms Section 4 compares them in. All four compute the score row through the same matrix-multiply sequence as the reference, in `ieee` precision: computing it as a reduction instead changes the summation order, and a rounding-level difference in a score is enough to change which peer is retained. The per-query kernel takes k argmax rounds over the score row and reads k value rows, so its aggregation is sparse and its selection is linear in k. The threshold kernel bisects on the score's integer bit pattern for 34 steps to find the k-th largest value exactly, then weights every peer by an indicator, so its selection is independent of k and its aggregation is dense. The sorted kernel maps each score to an order-preserving integer, packs it against its own peer index into one 64-bit word, sorts the row once, and reads the k value rows named by the top words, so it is independent of k and sparse at the same time. The pre-score kernel is given the retained indices and never forms a score row at all: one program per tile of queries walks that tile's retained peers, reading one key row and one value row per peer and accumulating with a running maximum, so nothing of size (B, N, k, d) is ever written.

B.3 Which backend serves the fused dense baseline

A probe forces each scaled-dot-product backend in turn, at both dtypes and at three agent counts, under no mask, an additive float mask and a Boolean mask, and the profiler records which kernel the default dispatcher launches. The FlashAttention backend is rejected under any non-null mask in either form, and in `fp32` under any input, so the library dense baseline runs the memory-efficient CUTLASS kernel. Forcing FlashAttention on the same unmasked `bf16` shape is faster than the kernel we do get, so a baseline able to reach it would widen rather than narrow the gap we report — which is part of why the dense baseline in this revision is the better of the library path and a kernel we wrote.

B.4 The host allocator control

Dense, masked and gathered each allocate and free several multi-MiB score matrices per forward. The default allocator returns those pages to the kernel and they are re-faulted on the next call, which at N=192 adds roughly 37,000 minor page faults per forward and inflates latency by up to 8.8×. The tax is proportional to the bytes of large temporaries each implementation churns, so it differs per implementation and does not cancel in a ratio; in an un-pinned run it inverted the ordering at one configuration. We raise the allocator's retention threshold so that freed temporaries are kept, and count minor faults per forward from the process's own resource usage across the timed region. The harness records the allocator environment, the thread counts, the affinity mask and the fault totals in each run's metadata. The fault figures quoted here come from the host microbenchmark that first exposed the effect, which predates the re-measurement campaign; the campaign's own artifacts inherit the fix and do not re-derive the fault counts, so those two numbers should be read as the diagnosis that motivated the control rather than as a released measurement.

B.5 Machine balance

Peak arithmetic rate is measured with a large matrix multiply and peak bandwidth with a large copy, in the same process as the measurements they normalise. That matters here more than it usually would: on this shared host the peak we measure varies between 58 and 77 TFLOP/s across sessions, depending on what else is running. Every fraction-of-peak figure in the paper is against the peak measured in the same process as the number it normalises, and is therefore an upper bound on the true efficiency rather than an estimate of it. Comparisons between two implementations measured in one process are unaffected; comparisons of a fraction across sections are not, and we make none. The copy source is written before it is read: an untouched allocation on this operating system is backed by the shared zero page, so a host-side copy from it re-reads one cache line and reports a bandwidth several times the machine's real figure, and an earlier version of this probe did exactly that: every CPU ridge point derived from it was too large by roughly a factor of six. Device work is also timed with device events rather than host wall-clock, since host timing around a sub-millisecond copy measures launch and synchronisation. The probe reports two ridge points per device, one with a cache-resident buffer and one with a buffer larger than the last-level cache, because the two differ by a factor of two on the host and a claim that holds for one need not hold for the other.

C Additional training analysis

C.1 Pre-registration chronology

The analysis plan was committed when eight of the forty runs in the pre-registered family had produced a final evaluation and before any of their returns had been inspected. Commit history establishes the ordering: the positive-control artifacts at 12:18, the plan at 12:56, and the remaining artifacts at 19:11 the same day. The plan has two commits, the second appending to its deviations log; the endpoint, comparisons, test, effect sizes, interval and correction are byte-identical to the pre-unblinding version. Timestamps establish ordering and nothing more. Data generation had demonstrably already begun.

The power statement in Section 6 is arithmetic, not an estimate. With five seeds per variant an exact two-sided permutation test over $\binom{10}{5} = 252$ assignments has smallest attainable p-value $2/252 = 0.0079$, against a first Holm threshold of $0.05/6 = 0.0083$: a margin of 3.97×10^{-4}. A first rejection therefore requires one variant's five seeds to lie entirely above the other's, and any effect smaller than complete separation is undetectable by construction.

One correction is logged against the released plan. It asserts that at most one comparison in the family can reach corrected significance. That is false: Holm is a step-down procedure, so once a first rejection is obtained its later thresholds are looser and further rejections remain possible. The manuscript states the corrected version; the deviation is logged against the released plan.

C.2 The remaining comparisons

Five further comparisons were registered and all are Holm-adjusted nulls. Two deserve a note. The comparison at twelve agents has the one sign opposite to its registered direction and carries the family's largest point estimate, and we set it aside on the ground that neither variant learns the task there. That ground was not registered: it is a post-hoc exclusion of the largest observed effect, we have logged it as a deviation against the released plan, and a reader who rejects it should read the comparison as an under-powered null with a large point estimate, which is what the registered analysis returns either way. The evidence for the exclusion is this. A uniform-random policy scores -758.9 under a forty-episode evaluation, against -864.9 for the dense variant, which is worse, and -717.9 for the sparse variant, which is better but barely learning; the comparison is between a diverged run and a struggling one. Separately, for two comparisons the bootstrap interval excludes zero while the permutation test does not. The registration named the permutation test, so it governs; the disagreement is unsurprising at five seeds, where the interval is coarse, discrete and of uncertain coverage. We report both so that the estimated effect and the strength of evidence for it are both visible.

Descriptively, and supporting no claim: the budgeted variant meets its budget but not by the intended route, its gate collapsing to an expected support near one within the first tenth of training while the dual multiplier decays to zero; the entmax variant settles at a support of 1.64 of five peers; and attention entropy falls from 0.35 nats over five peers to 0.12 over two. The three sit on different supports and are reported only as an instrumentation check.

C.3 The partially observable environment

The variant modifies the standard cooperative task in one respect: a peer's relative-position entry is zeroed when the peer is farther than a fixed radius. Reward, action space, landmark observations, episode length and observation dimension are unchanged. Two properties limit what it can support. The mask is a zeroing rather than an explicit visibility flag, so a peer at relative position exactly zero is not distinguishable from an absent one. On a continuous position space that coincidence has probability zero and we did not observe it, but we did not instrument for it either, and the released characterisation records visible fractions only. And the visible fraction falls with the number of agents while the absolute number visible rises, because the world size is fixed, which is a difficulty confound that rules out across-agent-count return comparisons.

The communicating variants carry between 7.2% and 19.0% more trainable parameters than the communication-free baseline, since the attention module adds parameters the baseline has no analogue for. The positive controls therefore establish that the communicating architecture outperforms the baseline, not that the channel alone accounts for the difference.

D Literature audit

For each audited paper we read the full text, including appendices and the official supplement where one exists — through the ar5iv rendering of the arXiv version where the paper is on arXiv, and the venue's own copy otherwise — searched it for seven temporal keywords, and recorded whether it makes a computational-efficiency or complexity claim about its mechanism, whether it reports any measured temporal quantity attributable to that mechanism, and the figure, table or sentence that settled the decision. A statement such as "training took a few days" does not qualify, since it is not attributable to the mechanism. Scope matters for at least one entry, whose affirmative verdict rests on a wall-clock figure in an appendix and would be negative on the main body alone. This is an audit of reporting practice in a small, deliberately selected set — the papers we ourselves relied on — and not a sample of either literature; a negative verdict means the argument rests on operation counts or asymptotics, not that it is wrong.

E Reproduction and provenance

The measurement campaign is training-free and completes in under an hour on one device. `scripts/run_revision_campaign.sh` regenerates the twelve artifacts every table and figure in

Table 13: The twelve audited papers, their efficiency claim, the kind of evidence it rests on, and whether any measured temporal quantity attributable to the mechanism appears anywhere in the paper or its supplement. A "no" means the argument rests on operation counts or asymptotics, not that it is wrong.

Paper	Efficiency claim	Type	Measured time?
Multi-agent communication and attention			
Das et al. (2019)	Qualitative bandwidth argument only	qualitative	no
Jiang & Lu (2018)	Communication is 'restricted by bandwidth . . .	agent/message counts	no
Jiang et al. (2020)	receiving a large amount of information . . .	asymptotic / qualitative	no
Singh et al. (2019)	continuous communication allows efficient . . .	message counts / convergence epochs	no
Sukhbaatar et al. (2016)	none explicit	—	n/a
Wen et al. (2022)	Sequence modelling 'reduces the complexity . . .	asymptotic	**yes**
Efficient attention and sparse mixture-of-experts			
Beltagy et al. (2020)	Sliding-window + global attention scales . . .	asymptotic	**yes**
Choromanski et al. (2021)	FAVOR+ gives linear rather than quadratic . . .	asymptotic	**yes**
Fedus et al. (2022)	Sparse routing increases parameters at . . .	FLOPs-matched	**yes**
Kitaev et al. (2020)	LSH attention reduces O(L2) to O(L log L)	asymptotic	**yes**
Shazeer et al. (2017)	1000x capacity increase at minor . . .	FLOPs + measured	**yes**
Zaheer et al. (2020)	reduces this quadratic dependency to linear'; . . .	asymptotic + memory	no

this paper is built from, in the order the manuscript reports them; the equivalence sweep (`scripts/verify_all_configurations.py`), the launch-cost probe (`scripts/launch_overhead.py`) and the backend probe of Appendix B are separate entry points and are named where they are used. Results not produced by the current campaign — the host-side allocator diagnosis and the numerical-format robustness checks — date from an earlier campaign on the same host, and we say so where they are quoted; `scripts/make_revision_tables.py` and `scripts/make_revision_figures.py` regenerate every table and figure from those artifacts, so no tabular number in the manuscript is transcribed by hand; and `scripts/check_revision_numbers.py` re-derives the numbers that appear in prose and fails on any disagreement. The campaign runs the tiling search first, because every sweep after it re-times the dense configuration the search selects; that is what makes each ratio in this paper a ratio against the best dense implementation we have rather than against whichever one a given sweep happened to build.

The environment is one NVIDIA RTX PRO 6000 Blackwell Server Edition (`sm_120`, 95 GiB, driver 610.43.02) of the two on a host with two Intel Xeon 6507P processors, under PyTorch 2.11 with CUDA 12.8 and Triton 3.6. Every latency in the paper is `fp32`, timed with CUDA events around a single call, interleaved implementation by implementation and repetition by repetition, with 20 warm-up repetitions discarded, 40 to 60 timed repetitions per block, and the whole block repeated three times; the reported figure is the minimum over all samples and the spread of the three per-run medians is recorded beside it. The tiling search enumerates query tiles of 16 to 128, key tiles or head chunks of 16 to 128, two to eight warps and one to three pipeline stages, and admits a configuration when its key, value and accumulator blocks fit a 100 kB shared-memory budget.

Three provenance caveats. The hardware description and the aggregate provenance record were written before this project's environment was upgraded and describe a superseded software stack and a host that is

no longer available; both have been corrected, and the run-level provenance for the training variants lives in the per-run configuration files rather than in the aggregate record. The GPUs used for every measurement in this paper are shared with other tenants, so absolute latencies are not reproducible to better than the drift reported in Appendix A.3 and only ratios between implementations timed in the same interleaved sweep should be compared. And the activation-memory measurement is reported from a run with the card otherwise free, since a shared-host repeat exhausts memory one agent count earlier.